

Permutations avoiding the vincular pattern 12–34

Part I: exact enumeration and complete asymptotics

Part II: a record and cycle bijection, uniform asymptotics,
and local large deviations

Part III: a diagonal-state route, block integrals
and non-D-finiteness

Part IV: exact Hankel sectors at $\log 4 + 2\pi i k$

(OEIS A113226)

AI-assisted research manuscripts 04 and 05 of batch 77
and Reports 87 and 94 of batch 101,
merged for the ProveIt research-report collection

Sources dated 1–2 October 2026; Parts I–II merged 2 October 2026;
Parts III–IV added 5 October 2026

Abstract

For OEIS A113226, the permutations avoiding the vincular pattern 12–34, equivalently the $\{3, 2 + 2\}$ -free naturally labelled posets of Bevan, Cheon and Kitaev, Part I derives a closed exponential generating function, proves the stretched-exponential form conjectured by those authors with exact constants and power $-5/6$, gives every algebraic correction in powers of $n^{-1/3}$ by a finite Gaussian algorithm, an exact quadratic-arithmetic recurrence and controlled Lambert- W inverse expansions. Part II marks the number of distinct nonempty strict downsets and of isolated elements: it explains the exponential transform behind A113226 by a record-to-cycle bijection, proves a complex-uniform all-orders saddle expansion of the marked generating function, precise all-orders local large deviations for interior densities, Gaussian and local limit laws, and a conditional Poisson law of isolated elements with an explicit correction and a total-variation error of order $n^{-2/3}$. All expansions are Poincaré expansions to each fixed order. The report is AI-assisted and unrefereed; no statement of it is formalized.

Added 5 October 2026 (batch 101). Parts III and IV print two further manuscripts. Part III is an independent, same-day second proof of Part I’s main theorem, by a diagonal-state argument directly on permutations; it adds integral forms of Elizalde’s block functions and proves that the generating function is differentially algebraic but not D-finite, so that A113226 is not P-recursive. Part IV writes $A_n/n!$ exactly as an absolutely convergent sum of Hankel sectors at the singularities $\log 4 + 2\pi i k$, isolates the first exponentially small conjugate pair with a geometric error, and gives every fixed sector a complete expansion; this answers Part I’s question on exponentially improved expansions for each fixed sector. Its sector expansions are again Poincaré expansions; the decomposition itself is exact.

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About this report

Sources and provenance

This report is built from **two** manuscripts of batch 77 (cluster P5), delivered in the arrival commit 096ee7b87 and placed in 4f11bc9c0. Both are dated 1 October 2026 and both record the same ProveIt pin, 63b9d6840 (their repository searches ran against that commit).

- **Source 04** (batch-77 manuscript 04, archive `a113226-asymptotics.zip`; *Exact enumeration and complete asymptotics for permutations avoiding 12 34*, 11-page PDF) is **Part I**: Sections 1–9 and Appendix A. It contributes the closed EGF, the ODE and cumulant recurrences, the identification with A136127, the singular and saddle analysis, the all-orders generator with $a_1, \dots, a_4$, the controlled inverse and the integer-threshold envelope.
- **Source 05** (batch-77 manuscript 05, archive `a113226-refined-addendum.zip`; *Refined enumeration of A113226: A record and cycle bijection, uniform asymptotics, and local large deviations*, 9-page PDF) is **Part II**: Sections 10–16. It is an addendum to source 04, written against the same pin; its delivery README says that the original A113226 report is unchanged. It contributes the downset/isolate marking, the record-to-cycle bijection, the complex-uniform first saddle, the marking phase gap, the local large deviations, and the Gaussian, local and Poisson laws.

The two sources form a version pair, not a supersession: Part II answers two of Part I’s further questions (see the write note after Part I’s list of questions in Section 9) and re-states Part I’s analysis at a general marking parameter u , which reduces to Part I at $u = 1$. Both are printed in full; no result, proof, remark, question or limitation of either source is omitted.

Where the merge had to choose. (i) The Parts are printed one after the other, not interleaved, because Part II’s derivations are the u -dependent versions of Part I’s and are easiest to check against them side by side; the $u = 1$ specializations are the ones Part II itself states. (ii) Part I keeps its delivered numbering exactly: Sections 1–9, equations (1)–(36), Theorem 1.1 and Appendix A. Part II’s Section n is Section $n + 9$ here and its equation (k) is equation $(k + 36)$; its Theorem 1, Lemma 1 and Theorem 2 are Theorem 12.1, Lemma 13.1 and Theorem 14.1 (theorems and lemmas share Part I’s section-based counter). (iii) The two bibliographies are merged; Part II’s citations of Bevan–Cheon–Kitaev, OEIS A113226 and A136127, Elizalde and Testart point to Part I’s entries for the same works, and its ProveIt entry is added as [8]. (iv) Three symbols of Part II are renamed because the source uses the same letter twice (Table 1). (v) Where Part II says “the original report” (four places) it now says “Part I” with a cross-reference. (vi) Write notes, set off and dated, record the relation to the sibling report, the repository context, the shipped file names and the status of Part I’s questions; no statement of either source was changed. (vii) The two large exact tables of the delivery are not shipped; the report README explains how to regenerate them or retrieve the delivered copies.

[Write note, 5 October 2026, batch 101.] **Parts III and IV.** Two further manuscripts, from the session bundle of batch 101 (arrival commit 60f54ea06, placement commit f7c612c72), are printed in full as Parts III and IV. *Report 87 (Exact Enumeration and Complete Asymptotics for Vincular Pattern Avoidance*, 1 October 2026, pin 1512ef835) is **Part III**, Sections 17–24: an independent answer to the same assignment as source 04, written the same day against an earlier repository state; it contributes a second proof of Part I’s theorem by a permutation-level diagonal state, integral forms of Elizalde’s block functions, and

Proposition 22.1 (differentially algebraic, not D-finite, not P-recursive). *Report 94 (Canonical Exponentially Small Sectors for Vincular Avoidance*, 2 October 2026, no pin; it depends on Report 87 by a SHA-256 of its archive) is **Part IV**, Sections 25–34 and Appendix B: the exact full-cut Hankel sector decomposition at $\rho + 2\pi ik$ and the complete expansion of every fixed sector, which answers Part I’s Question 4 for fixed k .

Where the addition had to choose. (viii) The two manuscripts are printed as Parts after Part II, not merged into Part I, because Part I is already written and Report 87’s proofs differ from it; each re-proof is marked as a second route with the Part I label it re-proves (see the note at the head of Part III). (ix) Part III’s Section n is Section $n + 16$ and its equation (k) is $(k + 67)$; Part IV’s Section n is Section $n + 24$ and its equation (k) is $(k + 91)$; Part IV’s Appendix A is Appendix B. Part I’s Appendix A keeps its letter and labels; its two unlabelled displays, which followed Part II’s equations as (68)–(69), now follow Part IV’s as (132)–(133). (x) One symbol of Part IV is renamed (the endpoint circle $C_k(r)$, see Table 3). (xi) The bibliographies are merged as for Part II; new entries are [9], [10], [11], [12] and [13]. (xii) Write results proved at this write are marked [write] (Remarks 19.1, 20.1, 27.3 and Proposition 23.2); unproved items of the two manuscripts are collected in Sections 24.1 and 34. An independent check of the write, made afterwards, is recorded at the end of Section 34; it corrected one sentence of Proposition 23.2 (dated note after its proof). No statement of either manuscript is changed.

Notation across the two Parts

Each Part keeps its own letters. Part II’s quantities depend on a marking parameter $u > 0$ (and s); at $u = s = 1$ they are Part I’s. Several letters nevertheless change meaning between the Parts, or within Part II; Table 1 lists them with the tempting false reading.

Relation to neighbouring work and status

Sibling report. The powered Catalan numbers, OEIS A113227, count the permutations avoiding the sibling vincular pattern 1–23–4. They are the subject of the sibling report

`SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/a113227-powered-catalan/`, built from two other batch-77 manuscripts (cluster P2, manuscripts 57 and 56). Elizalde’s 2006 paper [3] bounded the growth of avoiders of both patterns without a precise equivalent; the two reports supply the precise asymptotics for the two patterns, by unrelated methods (here an EGF and a stretched-exponential saddle; there a positive Bessel spectral representation). Neither pair of manuscripts cites the other (they are contemporaneous); Part I does cite Callan’s paper on the 1–23–4 pattern [7, Section 9], for the fast recurrence of Section 3.

Transseries volume. The Lambert- W inversion and the integer threshold envelope of Section 7 are instances of the inversion apparatus of the canonical transseries volume; see the write note at the end of that section. No novelty is claimed there.

Status. The report is AI-assisted and unrefereed. Its delivered review records (shipped beside this article; see the README) describe separate proof reviews and independent coefficient audits, and say themselves that they are not formal verification or conventional peer review. **No statement of this report is formalized**; its place in the research-report collection confers no formal status.

[Write note, 5 October 2026, batch 101.] The same holds for Parts III and IV: their delivered reviews and audits (shipped with prefixes 06-**diagonal**- and 07-**sectors**-) are AI-internal, the manuscripts are unrefereed, and nothing in them is formalized. Part III’s inversion is, like Part I’s, an instance of the transseries volume’s inversion apparatus; Part IV uses classical contour and saddle methods.

Part I

Exact enumeration and complete asymptotics (source 04)

Abstract of source 04. We derive a closed exponential generating function for OEIS A113226, the number of permutations avoiding the vincular pattern 12–34. The derivation uses the labelled binary-word model of Bevan, Cheon and Kitaev. The logarithm of the generating function satisfies a first-order linear differential equation and has an inverse-square-root singularity at $\rho = \log 4$. A saddle-point analysis proves their conjectured stretched-exponential form, with exact constants and power $-5/6$, and gives every algebraic correction in powers of $n^{-1/3}$. We provide an executable finite-order coefficient generator, the first four corrections, an exact quadratic-arithmetic enumeration recurrence, checks through $n = 1000$, and controlled Lambert- W inverse expansions. The known sequence A136127 is the logarithmic derivative’s coefficient sequence. The expansion is in the Poincaré sense; exponentially improved sectors remain a separate question.

1 The result and its context

Let A_n count permutations of $[n]$ avoiding the vincular pattern 12–34, with $A_0 = 1$, and put

$$H(z) = \sum_{n \geq 0} A_n \frac{z^n}{n!}, \quad L(z) = \log H(z), \quad L(0) = 0.$$

An occurrence requires positions $i, i+1, j, j+1$ with $i+1 < j$ and $\pi_i < \pi_{i+1} < \pi_j < \pi_{j+1}$. This adjacency convention matters. The sequence is OEIS A113226 [1].

Bevan–Cheon–Kitaev [2] identify these objects with $\{3, 2+2\}$ -free naturally labelled posets, give a labelled binary-word model and an insertion recurrence, and conjecture the asymptotic form

$$A_n \sim D n! \rho^{-n} \mu^{n^{1/3}} n^\beta, \quad \rho = \log 4.$$

Their Conjectures 13 and 14 specify the exponential rate and the stretched-exponential form, with fitted estimates for D, μ, β . Elizalde [3] obtained earlier bounds. The exact expression below resolves both conjectural forms and identifies their constants.

[Write note, 2 October 2026, batch 77P5.] Elizalde’s paper also bounds the avoiders of the sibling pattern 1–23–4 (the powered Catalan numbers, OEIS A113227) and leaves their precise asymptotics open. That question is answered, by a different method, in the sibling report `oeis-sequence-asymptotics/a113227-powered-catalan/` of this collection (batch 77, manuscripts 57 and 56); see “About this report”.

Define

$$c = \left(\frac{\pi^2}{4\rho} \right)^{1/3}, \quad D = 2e^{-3} \sqrt{\frac{c}{3\pi}}, \quad C = 3c, \quad \mu = e^C. \quad (1)$$

Theorem 1.1. *The exponential generating function is*

$$H(z) = \exp \left\{ \frac{z}{2} + \frac{q(3 \arcsin q - \pi/2)}{\sqrt{1-q^2}} \right\}, \quad q = \frac{e^{z/2}}{2}, \quad (2)$$

with branches analytic at $z = 0$. Its logarithm is equivalently the unique analytic solution of

$$(4 - e^z)L'(z) - 2L(z) = 2 + e^z - z, \quad L(0) = 0. \quad (3)$$

For every fixed integer $J \geq 0$,

$$A_n = D n! \rho^{-n} e^{3cn^{1/3}} n^{-5/6} \left(\sum_{j=0}^J a_j n^{-j/3} + O_J(n^{-(J+1)/3}) \right), \quad (4)$$

where $a_0 = 1$ and all a_j are given by the finite Gaussian coefficient formula in Section 6. In particular,

$$a_1 = \frac{c^2(1-\rho)}{2} - \frac{5}{36c}, \quad (5)$$

$$a_2 = \frac{324c^6(\rho-1)^2 + c^3(1908\rho - 612) - 35}{2592c^2}. \quad (6)$$

The constants are

$$\begin{aligned} c &= 1.21188506246461864249\dots, & D &= 0.03570604125804483339\dots, \\ C &= 3.63565518739385592747\dots, & \mu &= 37.92669385021515730\dots \end{aligned}$$

Thus $\beta = -5/6$. The paper’s decimal estimates are finite-range fits; they are not exact conjectured values. In particular, the amplitude D differs visibly from their estimate near 0.032.

The theorem is an all-orders Poincaré expansion. It neither asserts convergence of the coefficient series nor identifies all exponentially small contributions. Its combinatorial foundation belongs to [2]; the known sequence A136127 and its leading asymptotic are also prior work [4, 5, 6]. Our contribution is scoped to the exact A113226 transform, its complete coefficient asymptotics, and the resulting inverse.

[Write note, 2 October 2026, batch 77P5.] Part II extends Theorem 1.1 to the generating function marked by the number k of block pairs in (7) (the number of distinct nonempty strict downsets) and by the isolated elements, uniformly for the marking parameter in complex neighbourhoods of compact positive intervals (Theorem 12.1); at $u = 1$ it is the expansion (4).

[Write note, 5 October 2026, batch 101.] Part III (Report 87, same day, independent) proves Theorem 1.1 a second time, by a diagonal-state argument on permutations instead of the binary-word model (Theorems 17.1 and 17.2). The “exponentially small contributions” that the paragraph above leaves unidentified are identified in Part IV: $A_n/n! = \sum_k H_k(n)$ exactly, with contour-defined sectors at $\rho + 2\pi ik$ (Theorem 28.1); the sector H_0 has exactly the expansion (4) (divided by $n!$), and each other fixed sector has its own complete expansion (Theorem 30.1). Convergence of the coefficient series is still not asserted.

2 From labelled words to an integral equation

Proposition 7 of [2] gives a bijection with labelled binary words beginning in zero, whose labels decrease within a zero run, increase within a one run, and satisfy the condition that every one label exceeds all earlier zero labels. Equivalently, an object has the block form

$$E_1, M_1, E_2, M_2, \dots, E_k, M_k, E_{k+1}, \tag{7}$$

where E_i, M_i for $i \leq k$ are nonempty sets, E_{k+1} is an arbitrary set, and every label in M_i exceeds all labels in $E_1 \cup \dots \cup E_i$. Each block has its unique prescribed order. The case $k = 0$ includes the empty object and a single decreasing zero run.

Replace the integer labels by distinct coordinates in $[0, 1]$ and integrate over them. For fixed block cardinalities totaling n , divide by the factorial of every block size. Each ordering chamber has volume $1/n!$; the unordered coordinates inside each block supply exactly its block factorial. The coefficient is therefore the number of valid labelled objects divided by $n!$. There is no extra multinomial factor. Coincident coordinates have measure zero.

Let x be the largest coordinate among preceding zero blocks, initially $x = 0$. A new nonempty zero block has maximum at most x , with weight $e^{zx} - 1$, or creates a new maximum $t > x$, with density $ze^{zt} dt$. Indeed, the maximum of m coordinates has volume density mt^{m-1} , and summing $z^m mt^{m-1}/m!$ gives ze^{zt} . A following nonempty one block then has weight $e^{z(1-t)} - 1$. The final arbitrary zero block has weight e^z , independent of the state. Its labels have no restriction relative to earlier one labels.

If $f(x)$ counts any number of additional zero/one pairs, then

$$f(x) = 1 + (e^{zx} - 1)(e^{z(1-x)} - 1)f(x) + \int_x^1 z(e^z - e^{zt})f(t) dt. \quad (8)$$

This holds as a formal series, and analytically for small $|z|$. Each application of the operator increases size by at least two, so the formal solution is unique. Put

$$D_z(x) = e^{zx} + e^{z(1-x)} - e^z, \quad h(x) = D_z(x)f(x), \quad b(x) = z(e^z - e^{zx}).$$

Then

$$h(x) = 1 + \int_x^1 \frac{b(t)h(t)}{D_z(t)} dt, \quad h'(x) = -\frac{b(x)}{D_z(x)}h(x), \quad h(1) = 1.$$

Since $D_z(0) = 1$ and $H(z) = e^z f(0)$, this yields

$$L(z) = z + \int_0^1 \frac{z(e^z - e^{zx})}{D_z(x)} dx \quad (9)$$

$$= \frac{z}{2} + \frac{ze^z}{2} \int_0^1 \frac{dx}{D_z(x)}. \quad (10)$$

For the second equality average x and $1-x$ and use $e^{zx} + e^{z(1-x)} = D_z(x) + e^z$.

Set $v = z(x - 1/2)$ and $q = e^{z/2}/2$. The substitution $u = \tanh(v/2)$ in (10) gives

$$L(z) = \frac{z}{2} + \frac{2q}{\sqrt{1-q^2}} \arctan \left(\frac{2q-1}{2q+1} \sqrt{\frac{1+q}{1-q}} \right). \quad (11)$$

The arctangent vanishes at $q = 1/2$ and its derivative is $3/(2\sqrt{1-q^2})$. It therefore equals $(3/2) \arcsin q - \pi/4$ near $q = 1/2$. This proves (2). Differentiation proves (3).

3 Exact recurrences and the known cumulants

Let $\ell_n = L^{(n)}(0)$, with $\ell_0 = 0$. Coefficient extraction from (3) gives

$$3\ell_{n+1} = \sum_{k=0}^{n-1} \binom{n}{k} \ell_{k+1} + 2\ell_n + 1 + 2\mathbf{1}_{\{n=0\}} - \mathbf{1}_{\{n=1\}} \quad (n \geq 0). \quad (12)$$

The empty sum is zero. Since $H' = L'H$,

$$A_0 = 1, \quad A_{n+1} = \sum_{k=0}^n \binom{n}{k} \ell_{k+1} A_{n-k}. \quad (13)$$

These recurrences use $O(N^2)$ integer arithmetic operations for all terms through N . This statement counts arithmetic operations, not bit operations. It gives a fast recurrence of the kind asked for by Callan [7, Section 9].

The cumulants begin

$$(\ell_n)_{n \geq 1} = 1, 1, 2, 5, 16, 63, 294, 1585, 9692, 66275, \dots$$

They are the known sequence A136127 shifted by one:

$$\ell_n = \text{A136127}(n-1) \quad (n \geq 1). \quad (14)$$

Here is an exact verification that also explains the nonbinomial antidiagonal sum. The known poly-Bernoulli-relative array $T_{a,b}$ has bivariate exponential generating function

$$T(x, y) = \log \frac{1}{e^x + e^y - e^{x+y}} = \sum_{a,b \geq 1} T_{a,b} \frac{x^a y^b}{a! b!}.$$

The antidiagonal identity recorded in [6, Section 6] is

$$\text{A136127}(n) = \sum_{k=0}^{n-1} T_{k+1, n-k} \quad (n \geq 1), \quad \text{A136127}(0) = 1.$$

Equation (9) is exactly

$$L(z) = z + \int_0^z T_x(t, z-t) dt. \quad (15)$$

Since

$$\int_0^z \frac{t^{a-1} (z-t)^b}{(a-1)! b!} dt = \frac{z^{a+b}}{(a+b)!},$$

(14) follows, including $n = 1$ separately. Thus H'/H is the EGF of A136127. The existing objects, array and asymptotic for that sequence are not new claims of this report. A direct labelled bijection explaining the exponential transform remains an interesting question.

[Write note, 2 October 2026, batch 77P5.] Part II, Section 10.1, gives such an explanation: a record-to-cycle bijection whose components are “separated cycles”, with the component counts (39). It explicitly does not claim a direct bijection to a previously named exceedance-permutation or tree model of A136127, so that narrower question stays open (see the note after the questions in Section 9).

[Write note, 5 October 2026, batch 101.] Part III identifies L' with the EGF of A136127 by a second literature route, the Clark–Ehrenborg bivariate exceedance formula ((83)), and asks the same bijective question.

4 Analytic continuation and the singular expansion

For $|z| < \rho$, the elementary estimate

$$\begin{aligned} |(e^{zx} - 1)(e^{z(1-x)} - 1)| &\leq (e^{|z|x} - 1)(e^{|z|(1-x)} - 1) \\ &\leq (e^{|z|/2} - 1)^2 < 1 \end{aligned}$$

shows that (9) is analytic. The second inequality follows by minimizing $e^{rx} + e^{r(1-x)}$ at $x = 1/2$. On $|z| = \rho$, equality can occur only at $z = \rho, x = 1/2$. More strongly, the linear ODE continues L along every path avoiding

$$\rho + 2\pi i k, \quad k \in \mathbb{Z}.$$

In a disk of radius R with $\rho < R < |\rho + 2\pi i|$, cut along $[\rho, R)$, this continuation is single-valued and has no other singularities. This follows from analytic ODE existence and uniqueness on a simply connected domain.

[Write note, 5 October 2026, batch 101.] The points $\rho + 2\pi ik$ are genuine singularities of every continuation: by Part III's Proposition 22.1 each is an essential singularity on every branch, which also shows that the generating function is not D-finite. Part IV gives a global single-valued branch on $\mathbb{C}$ minus the horizontal cuts $\rho + 2\pi ik + [0, \infty)$ (period $4\pi i$, not $2\pi i$) and its exact local form at every $\rho + 2\pi ik$, with strength π for even and 2π for odd k ((98) and Lemma 26.1).

Write $w = \rho - z$, with the square root positive for real $w > 0$. Equation (2) gives a convergent local decomposition

$$L(z) = \frac{\pi}{\sqrt{e^w - 1}} + R(w), \quad (16)$$

where

$$R(w) = \frac{\rho - w}{2} - \frac{3 \arccos(e^{-w/2})}{\sqrt{e^w - 1}}$$

is analytic at zero: the square roots in the quotient cancel. Its Taylor expansion starts

$$R(w) = \frac{\rho}{2} - 3 + \frac{w}{2} - \frac{w^2}{10} - \frac{w^3}{210} + O(w^4). \quad (17)$$

All coefficients can be generated without repeated differentiation of inverse trigonometric functions. If $R(w) = \sum_{m \geq 0} r_m w^m$, then $r_0 = \rho/2 - 3$ and

$$r_m = \frac{-\mathbf{1}_{\{m=1\}} - 4(-1)^m/m! - 4 \sum_{k=1}^{m-1} \frac{(-1)^{m-k} k r_k}{(m-k+1)!}}{4m+2} \quad (m \geq 1). \quad (18)$$

Indeed substitution in (3) gives

$$4(1 - e^{-w})R'(w) + 2R(w) = \rho - w - 2 - 4e^{-w}.$$

Also define

$$s_m = [w^m] \sqrt{\frac{w}{e^w - 1}}, \quad s_0 = 1, \quad s_1 = -\frac{1}{4}, \quad s_2 = \frac{1}{96}, \quad s_3 = \frac{1}{384}.$$

Put $t = 1 - z/\rho$, $K = \rho/2 - 3$ and $A = \pi/\sqrt{\rho} = 2c^{3/2}$. Then, convergently near $\sqrt{t} = 0$,

$$L(\rho(1-t)) = K + At^{-1/2} + \sum_{j \geq 1} g_j t^{j/2}, \quad (19)$$

where

$$g_{2m-1} = \pi s_m \rho^{m-1/2}, \quad g_{2m} = r_m \rho^m \quad (m \geq 1). \quad (20)$$

In particular $g_1 = -\pi\sqrt{\rho}/4$ and $g_2 = \rho/2$. There is no logarithmic term. This fact determines the power $n^{-5/6}$ in the main result.

5 Contour estimates and a complete saddle expansion

Set $\delta = n^{-1/3}$, $t_0 = c\delta^2$, and take the Cauchy circle $z = \rho(1 - t_0)e^{i\theta}$. For fixed sufficiently small θ_0 , (19) gives uniformly for $|\theta| \leq \theta_0$

$$\log |H(z)| = K + A\Re(t^{-1/2}) + O(|t|^{1/2}), \quad t = 1 - (1 - t_0)e^{i\theta}. \quad (21)$$

For $|\theta| \leq \varepsilon t_0$, with ε small and fixed, the Taylor expansion of $(1+v)^{-1/2}$ yields

$$\Re(t^{-1/2}) \leq t_0^{-1/2} - b\theta^2 t_0^{-5/2} \quad (22)$$

for a constant $b > 0$. Here $\Re t \geq t_0$ and $\Im t$ is comparable to $-\theta$. For the complementary range $\varepsilon t_0 \leq |\theta| \leq \theta_0$, use

$$|t|^2 = t_0^2 + 2(1-t_0)(1-\cos\theta)$$

to obtain a fixed fractional loss

$$\Re(t^{-1/2}) \leq |t|^{-1/2} \leq t_0^{-1/2}(1+b'\varepsilon^2)^{-1/4}. \quad (23)$$

On the rest of the circle, L is uniformly bounded as n grows, by analytic continuation across the compact arc of $|z| = \rho$ that excludes ρ .

Consequently every arc outside

$$|\theta| \leq \delta^{5/2} \log(1/\delta) \quad (24)$$

is smaller than the main coefficient scale by every negative power of n . Estimate (22) treats the intermediate range; (23) treats the remaining local arc. The bounded error in (21) cannot offset either loss. The modulus of the Cauchy coefficient kernel is constant on this circle.

For a simple coefficient formula, deform the central arc to the vertical segment

$$t = c\delta^2(1 + i\sqrt{\delta}u). \quad (25)$$

One can choose endpoints with $|\Im t| = \varepsilon t_0$. They differ from the circular arc by $O(t_0^2)$ in real part. The connecting segments stay in $\Re t \geq t_0$, with $|\Im t|$ comparable to εt_0 , so the same fixed fractional loss applies. Their change in $-n \log |1-t|$ is $O(nt_0^2) = O(\delta)$. They are exponentially negligible, and no singularity is crossed. Reversing orientation gives a positive Gaussian integral.

The leading phase on (25) is

$$At^{-1/2} + nt = \frac{3c}{\delta} - \frac{3c}{4}u^2 + O(\sqrt{\delta}|u|^3). \quad (26)$$

The Cauchy Jacobian is $c\delta^{5/2}/(2\pi)$; the remaining $(1-t)^{-1}$ will be included explicitly below. Therefore the leading factor, apart from $n!\rho^{-n}e^{3c/\delta}$, is

$$\frac{e^K c\delta^{5/2}}{2\pi} \sqrt{\frac{4\pi}{3c}} = 2e^{-3} \sqrt{\frac{c}{3\pi}} n^{-5/6}. \quad (27)$$

In particular $nt^2/2 = O(\delta)$ affects the first correction, not this constant.

6 A finite generator for every correction

Let U be a centered Gaussian variable of variance $2/(3c)$. Define the following formal series in $\sqrt{\delta}$:

$$\begin{aligned} Q(\delta, U) &= 2c \sum_{k \geq 3} \binom{-1/2}{k} (iU)^k \delta^{k/2-1} \\ &\quad + \sum_{m \geq 2} \frac{c^m}{m} \delta^{2m-3} (1 + i\sqrt{\delta}U)^m \end{aligned}$$

$$\begin{aligned}
& + \sum_{m \geq 1} \frac{c^m}{m} \delta^{2m} (1 + i\sqrt{\delta}U)^m \\
& + \sum_{j \geq 1} g_j c^{j/2} \delta^j (1 + i\sqrt{\delta}U)^{j/2}.
\end{aligned} \tag{28}$$

The four lines are, respectively, the principal phase after removing its constant, linear and quadratic terms; $n[-\log(1-t) - t]$; the extra $-\log(1-t)$; and the Puiseux perturbation. The coefficients in Theorem 1.1 are

$$a_j = [\delta^j] \mathbb{E} e^{Q(\delta, U)}. \tag{29}$$

Every requested coefficient involves finitely many terms. Odd powers of $\sqrt{\delta}$ are odd in U and integrate to zero. The Gaussian moments then imply $a_j \in \mathbb{Q}[c, c^{-1}, \rho]$.

We spell out why the formal calculation supplies the asserted error bound. Restrict (25) to $|u| \leq \log(1/\delta)$. Its complement is superpolynomially small by the quadratic estimate. On this window the principal phase remainder is $O(\sqrt{\delta}(1 + |u|^3))$. Every other summand and every fixed finite Taylor remainder are bounded by a power of δ times a polynomial in u , uniformly on the window. This follows from the convergent local expansion (19) and the analytic binomial and logarithm series. The exact integrand after its leading normalization is bounded by $C_0 e^{-b_0 u^2}$, since $\sqrt{\delta}|u|^3 = o(1 + u^2)$ there. Expanding to two extra half-orders, and bounding the exponential Taylor remainder by an integrable polynomial times $e^{-b_0 u^2}$, gives a remainder $O_J(\delta^{J+1})$. The next odd half-order vanishes on the symmetric window and on its Gaussian extension. No residual logarithmic loss is hidden in the remainder. This completes the proof of (4) and (29).

For example the coefficient of δ is

$$\frac{c^2(1-\rho)}{2} + \frac{35c}{64} \mathbb{E} U^4 - \frac{25c^2}{128} \mathbb{E} U^6 = \frac{c^2(1-\rho)}{2} - \frac{5}{36c}.$$

The first four values are

j	a_j
1	-0.39827424329477706556...
2	0.98158861254222799575...
3	-0.53022033843535145958...
4	-0.66858737252587671304...

Exact expressions for a_3, a_4 are given in Appendix A.

The accompanying `asymptotic_coefficients.py` implements the finite algorithm for arbitrary requested fixed order J . It generates s_m and (18), constructs (28) to order h^{2J} with $h = \sqrt{\delta}$, and computes

$$E_0 = 1, \quad E_k = \frac{1}{k} \sum_{j=1}^k j Q_j E_{k-j}, \quad Q = \sum Q_j h^j, \quad e^Q = \sum E_k h^k. \tag{30}$$

It replaces U^{2m} by $(2m-1)!![2/(3c)]^m$ and odd powers by zero. Default execution computes $J = 4$. Larger orders may incur significant symbolic time and memory cost. The stored decimal constants are high-precision evaluations, not certified interval enclosures.

7 A controlled inverse and integer thresholds

There exists a smooth increasing realization $F(x)$ for all sufficiently large x with $F(n) = A_n$, whose logarithm has a differentiable all-orders expansion. Here is a construction, to avoid assuming a canonical analytic interpolation. Realize the formal logarithmic expansion by a locally finite sum with successively later smooth cutoffs. Choose the cutoffs so its tail and each fixed derivative satisfy the desired bounds. If G is the resulting smooth logarithm, the discrepancies $\log A_n - G(n)$ are smaller than every negative power of n . Multiply each discrepancy by a fixed smooth bump centered at n , with value one there and support of radius less than $1/3$. The bumps have disjoint supports; their sum and each fixed derivative are flat to all algebraic orders. Exponentiating the corrected G gives F . Its logarithmic derivative is asymptotic to $\log(x/\rho) > 0$, proving eventual monotonicity.

Write

$$\log F(x) = x \log \frac{x}{e\rho} + Cx^{1/3} + \alpha \log x + d + \sum_{j \geq 1} \beta_j x^{-j/3}, \quad (31)$$

where

$$\begin{aligned} C &= 3c, & \alpha &= -\frac{1}{3}, & d &= \log(D\sqrt{2\pi}), \\ \beta_1 &= a_1, & \beta_2 &= a_2 - \frac{a_1^2}{2}, & \beta_3 &= a_3 - a_1 a_2 + \frac{a_1^3}{3} + \frac{1}{12}. \end{aligned}$$

The $1/12$ comes from Stirling's expansion for $\log \Gamma(x+1)$; later Stirling terms are included at their respective orders.

For $Y \rightarrow \infty$ define

$$L_Y = \log Y, \quad N = \frac{L_Y}{W(L_Y/(e\rho))}, \quad \ell = \log(N/\rho) = W(L_Y/(e\rho)) + 1. \quad (32)$$

Thus $N \log(N/(e\rho)) = L_Y$. Put

$$a = -\frac{C}{\ell}, \quad b = \frac{1}{3} + \frac{\log \rho/3 - d}{\ell}, \quad (33)$$

$$d_1 = -\frac{a^2/2 + Ca/3 + \beta_1}{\ell}, \quad e_1 = -\frac{(a + C/3)b + \alpha a + \beta_2}{\ell}. \quad (34)$$

Then

$$F^{-1}(Y) = N + N^{1/3}a + b + N^{-1/3}d_1 + N^{-2/3}e_1 + O\left(\frac{1}{N\ell}\right). \quad (35)$$

Omitting the two decaying corrections leaves an error $O(N^{-1/3}/\ell)$, already $o(1)$.

To prove (35), substitute its displayed approximation in (31). The terms of orders $N^{1/3}$, 1 , $N^{-1/3}$ and $N^{-2/3}$ vanish successively by (33)–(34). The remaining logarithmic residual is $O(N^{-1})$. Its derivative is asymptotic to ℓ , so the mean value theorem divides that residual by a quantity comparable to ℓ . All four coefficient functions remain bounded as $\ell \rightarrow \infty$.

Successive cancellation gives a complete inverse expansion in powers of $N^{-1/3}$, beginning with $N^{1/3}$, with coefficients rational in ℓ . There is no separate growing logarithm in these coefficients, since $\log N = \ell + \log \rho$. Every logarithmic residual is converted to an inverse error by the same derivative bound. Changing the smooth realization by a flat function leaves every algebraic order unchanged.

For the eventual integer threshold $\tau(Y) = \min\{n : A_n \geq Y\}$, the monotone realization gives $\tau(Y) = \lceil F^{-1}(Y) \rceil$ for sufficiently large Y . If an approximation X has an established bound $|F^{-1}(Y) - X(Y)| \leq E(Y)$, then

$$\lceil X - E \rceil \leq \tau(Y) \leq \lceil X + E \rceil. \quad (36)$$

An $o(1)$ approximation alone does not justify unconditional rounding when X approaches an integer. The numerical checks below are not certificates of explicit constants in the asymptotic remainder.

[Write note, 2 October 2026, batch 77P5.] This section re-derives, for A113226, the inversion apparatus of the canonical transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`: the exact Lambert- W solution of the dominant block (`p0:thm:lambert-core`) behind (32), the perturbed inversion (`p0:thm:perturbed-inversion`) behind (35), and the staircase theorem (`p0:thm:staircase`), whose exact-rounding identity is $\tau(Y) = \lceil F^{-1}(Y) \rceil$ and whose separation clause is the use of the envelope (36). No novelty is claimed for that apparatus. The separation clause, and its failure for arbitrarily small error, are formalized in Lean as `Fabius.staircase_separation` and `Fabius.staircase_separation_fails` in `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`; these are generic real-number statements, and nothing specific to A113226 is formalized.

[Write note, 5 October 2026, batch 101.] Part III's Section 23 gives a variant: rigorous ceiling brackets for the finite models f_K (Theorem 23.1), without a smooth realization, and a reversion in the gamma core $\log \Gamma(X+1) - X \log \rho$ instead of the Lambert- W core; its displayed two-term reversion is proved in Proposition 23.2 [write]. Both are again instances of the transseries volume's apparatus.

8 Exact checks and reproducibility

The file `exact_recurrence.py` computes (12) and (13) using integer arithmetic. The released data contain all terms through $n = 1000$. All 21 displayed OEIS values agree. Two independent combinatorial checks are included:

- The five insertion rules in Proposition 11 of [2] agree through $n = 65$. This implementation does not use the EGF or cumulants.
- Exhaustive permutation enumeration with the literal vincular pattern predicate agrees through $n = 8$.

These checks supplement the combinatorial proof; they do not replace it.

Let R_n be the ratio of A_n to the leading term in (4), and let

$$T_j(n) = n^{(j+1)/3} \left(R_n - \sum_{k=0}^j a_k n^{-k/3} \right).$$

The theorem predicts $T_j(n) \rightarrow a_{j+1}$. Some exact-data comparisons are shown below. The convergence is slow, consistent with the $n^{-1/3}$ correction scale.

n	R_n	$T_0(n)$	$T_1(n)$	$T_3(n)$
100	0.9535673415	−0.2155213093	0.8482639777	−0.4113273816
200	0.9575584055	−0.2481999506	0.8776397878	−0.4542524517
400	0.9625345125	−0.2760480719	0.9005701306	−0.4916630397
557	0.9650312573	−0.2877167077	0.9096480952	−0.5076047828
1000	0.9694049550	−0.3059504504	0.9232379289	−0.5328649788
Limit	1	$a_1 = -0.3982742433$	$a_2 = 0.9815886125$	$a_4 = -0.6685873725$

At $n = 1000$, the scaled remainder $T_4(n)$ is about 1.3572. The data are consistent with the next algebraic correction, but this comparison does not certify a uniform remainder bound.

At $Y = A_{1000}$ the inverse approximation errors after the constant term b , after adding $N^{-1/3}d_1$, and after adding $N^{-2/3}e_1$ are respectively

$$-0.0115393\dots, \quad +0.00233040\dots, \quad -0.0000291627\dots$$

All sampled ratios and inverse errors are stored in `validation.json`.

The reproducibility archive contains the report source, exact data, Python programs, pinned dependencies, a build script, a replay script, and review/validation records. A separate proof review and independent hand-expanded Gaussian checks of a_1, a_2 found no unresolved defect. These checks are not formal verification or conventional peer review. A short replay regenerates the asymptotic coefficients and numerical comparison file, while checking a smaller exact prefix against the supplied $n = 1000$ data. Full exact regeneration is separately available. No network access is needed for the mathematics once the standard dependencies are installed.

[Write note, 2 October 2026, batch 77P5.] In this collection the programs and records of this section are shipped under prefixed names: `code/04-asymptotics-exact_recurrence.py`, `code/04-asymptotics-asymptotic_coefficients.py`, `code/04-asymptotics-validate_asymptotics.py`, `code/04-asymptotics-replay.sh`, `data/04-asymptotics-validation.json` and so on, with the proof revision and the reviews at the report root (`04-asymptotics-proof.md`, `04-asymptotics-root-mathematical-review.md`, `04-asymptotics-root-prior-art-check.md`). The exact data through $n = 1000$ (`exact_values.json`, 2,236,789 bytes) are *not* shipped: they are regenerated byte-identically by the exact recurrence program, and the delivered copy survives in the arrival commit; the report README gives the commands. The delivered PDF, manifest and build output are not shipped either; this article replaces the delivered source.

9 Prior work and further questions

The focused screening inspected the current OEIS entry, the Bevan–Cheon–Kitaev preprint and journal record, Elizalde’s primary paper, A136127 and its listed literature, and Testart’s 2026 paper. The exact A113226 EGF and logarithmic-derivative relation were not found in those sources. This is a scoped priority check, not an exhaustive claim about all literature. The underlying binary-word model, its bijections, the cumulant sequence and standard saddle methods are explicitly credited.

A contemporaneous default-branch search of the public ProveIt repository for A113226, 12–34, and the author names returned no matches. The checked head was [63b9d68407f2](#) (full hash in the source-check record). The original source model and conjectures remain independently verifiable through the references below.

[Write note, 2 October 2026, batch 77P5.] The search above ran against the pin 63b9d6840, which is kept here as provenance. The repository now contains this report (placed in 4f11bc9c0) and the sibling report on A113227, `oeis-sequence-asymptotics/a113227-powered-catalan/`; no other report of the repository treats A113226, A136127 or the BCK model. The “source-check record” is shipped as `data/04-asymptotics-repository-check.json`.

Several extensions are meaningful but are not needed for the theorem:

1. Find a direct labelled bijection explaining why A136127 is the logarithmic derivative’s coefficient sequence.
2. Refine the transfer equation by statistics such as the number of zero/one blocks or natural poset parameters, and determine their limiting laws.
3. Establish certified numerical constants in finite- n error bounds and threshold envelopes.
4. Develop exponentially improved expansions accounting for continuation near the other singularities $\rho + 2\pi ik$.

The last question is distinct from the complete algebraic expansion proved here.

[Write note, 2 October 2026, batch 77P5.] Status of these questions in this report. *Question 1* is answered by Part II, Section 10.1: separated cycles and a record-to-cycle bijection explain the exponential transform; a direct map to a particular previously named A136127 model (excedance permutations, non-ambiguous trees) is not claimed and remains open. *Question 2* is answered by Part II for the number of block pairs (distinct nonempty strict downsets) and the number of isolated elements: exact bivariate marking (37)–(38), local large deviations (Theorem 14.1) for densities in compact subsets of $(0, 1/2)$, Gaussian and local limit laws, and Poisson laws of isolated elements. Endpoint densities ($k = o(n)$ or $n/2 - k = o(n)$) and other natural poset parameters remain open. *Questions 3 and 4* remain open; Part II does not address certified constants, and it also excludes the other complex singularity sectors.

[Write note, 5 October 2026, batch 101.] Update for Parts III–IV. *Question 4 is answered for every fixed sector by Part IV*: the exact decomposition $A_n/n! = \sum_k H_k(n)$ (Theorem 28.1), the first exponentially small pair with an absolute error (131), and complete expansions of each fixed sector (Theorem 30.1). Still open, and listed in Section 34: expansions uniform in a growing sector index, the growth of a_j and optimal truncation against the exact H_0 (no Borel summability or resurgence is claimed), and lower bounds for the oscillating pair. *Question 3* remains open; Part IV’s “certified numerical extraction” is a case of it. *Question 1* is asked again by Part III, with the same status (open in its narrow form). On *Question 2*, Part III asks for limit laws of the number of components of the exponential formula, which Part II does not mark (Section 24.1).

Part II

Refined enumeration (source 05)

[Write note, 2 October 2026, batch 77P5.] This Part prints source 05, an addendum to source 04 written against the same repository pin. Its Section n is Section $n + 9$ here and its equation (k) is equation ($k + 36$). “The original report” of the source is Part I. Its expanded

companion proof is shipped as 05-refined-refined-proof.md; its reviews and literature screen as 05-refined-mathematical-review.md, 05-refined-root-refinement-review.md, 05-refined-reproduction-review.md, 05-refined-typesetting-review.md and 05-refined-literature-screen.md. These delivered records cite the delivered nine-page PDF's equation numbers, which are the numbers above minus 36.

Abstract of source 05. We refine the enumeration of $\{3, 2+2\}$ -free naturally labelled posets by their number of distinct nonempty strict downsets and their isolated elements. A record-to-cycle bijection directly explains the exponential transform behind A113226. Its marked generating function admits a complex-uniform, all-orders saddle expansion. A second saddle gives precise local large deviations throughout the interior density range, including the correction caused by the stretched exponential. Gaussian and local limit laws follow, and isolated elements are asymptotically Poisson even conditionally on an atypical downset density, with an explicit correction and a total-variation error of order $n^{-2/3}$. All expansions are Poincaré expansions to each fixed order; exponentially complete transseries are not claimed.

10 Exact marking and a direct bijection

Let $A_{n,k,j}$ count $\{3, 2+2\}$ -free naturally labelled posets on $[n]$ having k distinct nonempty strict downsets and j isolated elements. The total is A113226(n) through the bijection of Bevan–Cheon–Kitaev (BCK) with 12–34-avoiding permutations [2]. The statistics here are defined on the poset side; an identification with a usual permutation statistic is not asserted.

BCK Proposition 7 writes a poset as

$$E_1, M_1, \dots, E_k, M_k, E_{k+1}.$$

The first $2k$ blocks are nonempty; the last is arbitrary and consists precisely of the isolated elements. The M_i group maxima with equal nonempty downsets, and the E_i are successive differences of these nested downsets. Each label of M_i exceeds every label of $E_1 \cup \dots \cup E_i$. Zero blocks are written in decreasing order and one blocks in increasing order.

Put

$$A_n(u, s) = \sum_{k,j} A_{n,k,j} u^k s^j, \quad H(z, u, s) = \sum_{n \geq 0} A_n(u, s) \frac{z^n}{n!}, \quad H(z, u) = H(z, u, 1).$$

The marked transfer identity is

$$H(z, u, s) = e^{(s-1)z} H(z, u), \tag{37}$$

$$\log H(z, u) = z + \int_0^1 \frac{uz(e^z - e^{zx})}{1 - u(e^{zx} - 1)(e^{z(1-x)} - 1)} dx. \tag{38}$$

To obtain it, apply the maximum-coordinate transfer of Part I (Section 2), putting weight u on every added block pair. With

$$a(x) = (e^{zx} - 1)(e^{z(1-x)} - 1), \quad b(x) = z(e^z - e^{zx}),$$

the continuation function satisfies

$$f(x) = 1 + ua(x)f(x) + u \int_x^1 b(t)f(t) dt.$$

For $h = (1 - ua)f$, differentiation gives $h' = -ubh/(1 - ua)$ and $h(1) = 1$. The final block supplies e^{sz} . Integration over label-order chambers gives the coefficient normalization $1/n!$.

10.1 The record and cycle construction

A *separated cycle* is a cyclic list of pairs (E_i, M_i) of nonempty disjoint label sets such that every E label is smaller than every M label. Cycles are identified under rotation, never reflection. Each has a unique representative beginning with the E block containing its largest E label.

Given a BCK word, remove its final isolated block. Set $e_i = \max E_i$ and cut immediately before every strict left-to-right record of $e_1, \dots, e_k$. In a segment beginning at r , the first maximum e_r exceeds all later E -block maxima in that segment. Every M label in the segment exceeds e_r by the original word condition. The segment therefore represents a separated cycle in its unique maximum- E -first rotation. Retain the cycles as an unordered set, and the isolated labels as singleton components.

Conversely, rotate each separated cycle to that representative, sort the cycles by increasing largest E label, and concatenate them. Every M label exceeds all E labels in its own cycle. Its cycle maximum exceeds every E label of all earlier cycles, so the cross-cycle inequalities hold as well. Append the singleton labels as the final decreasing zero block. Each cycle start is an E -maximum record, and no later pair within that cycle is a record. The cuts recover exactly the original cycles. These operations are inverse, including empty objects, and preserve both statistics.

Let $\left\{ \begin{smallmatrix} n \\ m \end{smallmatrix} \right\}$ denote a Stirling number of the second kind and let $L_n(u) = n![z^n] \log H(z, u)$. We have $L_1 = 1$ and, for $n \geq 2$,

$$L_n(u) = \sum_{m=1}^{\lfloor n/2 \rfloor} m!(m-1)!u^m \sum_{a=m}^{n-m} \left\{ \begin{smallmatrix} a \\ m \end{smallmatrix} \right\} \left\{ \begin{smallmatrix} n-a \\ m \end{smallmatrix} \right\}. \quad (39)$$

Indeed the a labels assigned to the E blocks of one component must be its a smallest labels. Partition its two label sets into m blocks, match the blocks in $m!$ ways, and cyclically order the pairs in $(m-1)!$ ways. The labelled set construction then gives

$$A_0(u, 1) = 1, \quad A_{n+1}(u, 1) = \sum_{r=0}^n \binom{n}{r} L_{r+1}(u) A_{n-r}(u, 1). \quad (40)$$

Thus the first polynomials are $1, 1, 1+u, 1+5u, 1+17u+5u^2$. At $u=1$ the component counts, including the singleton, are the shifted A136127 sequence identified in Part I, (14). This provides explicit cycle objects for the exponential transform; a direct bijection to a particular previously named excedance-permutation or tree model is not claimed.

For an analytic check of (39), set

$$T(x, y, u) = -\log(1 - u(e^x - 1)(e^y - 1)).$$

Then (38) is $z + \int_0^z T_x(t, z-t, u) dt$. Expansion of T and a beta integral give the same formula.

11 A convergent local singular expansion

For $u > 0$, define

$$a = u^{-1/2}, \quad \rho = 2 \log(1+a), \quad c = \left(\frac{\pi^2 a}{4\rho} \right)^{1/3}, \quad K = \frac{\rho}{2} - 2 - a, \quad D = e^K \sqrt{\frac{c}{3\pi}}. \quad (41)$$

These functions extend holomorphically to a sufficiently small complex neighborhood of each compact positive u -interval, using the positive real branches. With $w = \rho - z$,

$$\log H(\rho - w, u) = \pi\sqrt{a}w^{-1/2}S(w, u) + R(w, u), \quad (42)$$

where S, R are jointly analytic near $w = 0$, $S(0, u) = 1$, and $R(0, u) = K$. The following formulas determine every coefficient:

$$\begin{aligned} q &= \cosh(w/2) - a \sinh(w/2), & h &= a \cosh(w/2) - \sinh(w/2), \\ \mathcal{T} &= \coth((\rho - w)/4), & S &= \frac{h}{a} \sqrt{\frac{aw}{1 - q^2}}, \\ R &= \frac{\rho - w}{2} - \frac{2h}{1 + q} \sum_{m \geq 0} \frac{(-1)^m \mathcal{T}^{2m+1}}{2m + 1} \left(\frac{1 - q}{1 + q} \right)^m. \end{aligned} \quad (43)$$

The quotient in S is analytic and equals one at the origin; the series for R converges normally in a common sufficiently small neighborhood.

For derivation, symmetrize (38). With $q = \cosh(z/2) - 1/(2ue^{z/2})$ and $N = 1 - u + ue^z$, the elementary integral is

$$\log H = \frac{z}{2} + \frac{N}{ue^{z/2}\sqrt{1 - q^2}} \arctan \left(\tanh(z/4) \sqrt{\frac{1 + q}{1 - q}} \right).$$

Near the positive singularity use $\arctan X = \pi/2 - \arctan(1/X)$ and $N/(ue^{z/2}) = 2h$. This yields (42)–(43). Apparent $q = -1$ points elsewhere are not presumed singular; some are removable.

Writing $t = 1 - z/\rho$, we obtain the convergent expansion

$$\log H(\rho(1 - t), u) = K + 2c^{3/2}t^{-1/2} + \sum_{j \geq 1} g_j t^{j/2}. \quad (44)$$

If $S = \sum s_m w^m$ and $R = \sum r_m w^m$, then

$$\begin{aligned} g_{2m-1} &= \pi\sqrt{a} s_m \rho^{m-1/2}, & g_{2m} &= r_m \rho^m, \\ s_1 &= \frac{a^2 - 3}{8a}, & s_2 &= \frac{9a^4 + 10a^2 - 15}{384a^2}, & r_1 &= \frac{4 - a^3}{6a}. \end{aligned} \quad (45)$$

These specialize at $u = 1$ to Part I's expansion (19)–(20).

12 Complex uniformity of the first saddle

Theorem 12.1. *For each compact interval $\mathcal{U} \subset (0, \infty)$ there is a complex neighborhood Ω of $\mathcal{U}$ and holomorphic coefficients $a_j(u)$, $a_0 = 1$, such that for every fixed $J \geq 0$,*

$$\frac{A_n(u, 1)}{n!} = D\rho^{-n}n^{-5/6}e^{3cn^{1/3}} \left(\sum_{j=0}^J a_j(u)n^{-j/3} + O_{\mathcal{U}, J}(n^{-(J+1)/3}) \right), \quad (46)$$

uniformly on a fixed closed neighborhood of $\mathcal{U}$ contained in Ω . The normalized remainder is holomorphic and has the same fixed-derivative bounds on a smaller neighborhood. Uniformly for s in any fixed compact subset of $\mathbb{C}$, the analogous result has leading amplitude $De^{(s-1)\rho}$ and computable coefficients $a_j(u, s)$.

Proof. We supply the complex-parameter details because real pointwise asymptotics would be insufficient for the later limit and local laws. For real $u \in \mathcal{U}$, $r = \rho(u)$ and $|z| \leq r$,

$$|u(e^{zx} - 1)(e^{z(1-x)} - 1)| \leq u(e^{|z|x} - 1)(e^{|z|(1-x)} - 1) \leq 1. \quad (47)$$

On $|z| = r$, equality is possible only at $x = 1/2, z = r$. For $0 < x < 1$, equality in the exponential power-series triangle inequality forces z positive real; the endpoints $x = 0, 1$ give zero. Thus the denominator in (38) is nonzero on the compact normalized set $z = \rho(u)\zeta$, $|\zeta| \leq 1$, $|\zeta - 1| \geq \epsilon$. It remains nonzero in an open neighborhood of this set for u in a small complex neighborhood of $\mathcal{U}$. Near $\zeta = 1$, the jointly analytic local representation (42) supplies continuation in a common slit neighborhood. The two representations agree on their initial intersection. This constructs the needed common dented domain, without classifying every apparent singularity of the elementary expression.

Set $\delta = n^{-1/3}$ and use the circle

$$z = \rho(u)(1 - c(u)\delta^2)e^{i\theta}, \quad -\pi \leq \theta \leq \pi. \quad (48)$$

Shrink Ω so that $\Re c > c_* > 0$, $\Re(1/c) > b_* > 0$, and all the continuation and local expansion bounds hold uniformly. Near $\theta = 0$, $\Re(1 - (1 - c\delta^2)e^{i\theta}) > 0$ for large n , so the circle lies in the common domain. On its fixed outer arc $\log H$ is bounded, whereas the main exponential modulus is $e^{3\Re(c)/\delta}$ and the radial Cauchy factor is $e^{\Re(c)/\delta + O(\delta)}$.

On the local arc put $\theta = \delta^2 y$. Up to $O(\delta)$ for bounded y , the nonconstant leading phase, after removing the radial factor, is $\delta^{-1}\Phi_c(y)$, where

$$\Phi_c(y) = 2c^{3/2}(c - iy)^{-1/2} - iy, \quad \Phi_c(0) = 2c, \quad \Phi'_c(0) = 0, \quad \Phi''_c(0) = -\frac{3}{2c}.$$

For small $|y|$ the real-part loss is at least by^2/δ , up to a harmless small shift. More precisely the exact circle leaves an uncanceled linear term $O(\delta^{-1}\theta) = O(\delta|y|)$. On the Gaussian scale $y = \sqrt{\delta}X$ it is $O(\delta^{3/2}|X|)$; completing the square gives a negligible shift. Outside a growing Gaussian cutoff it is absorbed by the quadratic loss.

For $\epsilon \leq |y| \leq M$, there is a uniform fixed loss at real u , since $\Re(c - iy)^{-1/2} < c^{-1/2}$, and it persists under small complex perturbations by compactness. If $|\theta| \geq M\delta^2$ within the local arc, then $|t| \geq b|\theta|$ and

$$|2c^{3/2}t^{-1/2}| \leq C\delta^{-1}/\sqrt{M}.$$

Choose M large. This gives a fixed fractional loss from the main exponent. Hence every arc outside $|\theta| \leq \delta^{5/2} \log(1/\delta)$ is smaller than the main scale by every power of n , uniformly in u . Bounded $O(t^{1/2})$ perturbations do not change the conclusion.

Deform the central integral to

$$t = c\delta^2(1 + i\sqrt{\delta}X), \quad X \in \mathbb{R}. \quad (49)$$

For real u , take short connecting curves at fixed scaled endpoints $|X| = \epsilon/\sqrt{\delta}$, inside open neighborhoods with a fixed leading-phase loss. Continue this choice of curves continuously as u leaves the real interval, using a finite cover of $\mathcal{U}$. Their compact images remain in the loss region after shrinking the common neighborhood. No connector crosses $t = 0$ or the cut. This specifies a uniform valid deformation for complex u .

On $|X| \leq \log(1/\delta)$ the phase is $3c/\delta - 3cX^2/4 + Q$, with

$$\begin{aligned} Q = & 2c \sum_{m \geq 3} \binom{-1/2}{m} (iX)^m \delta^{m/2-1} \\ & + \sum_{m \geq 2} \frac{c^m}{m} \delta^{2m-3} (1 + i\sqrt{\delta}X)^m + \sum_{m \geq 1} \frac{c^m}{m} \delta^{2m} (1 + i\sqrt{\delta}X)^m \\ & + \sum_{j \geq 1} g_j c^{j/2} \delta^j (1 + i\sqrt{\delta}X)^{j/2}. \end{aligned} \quad (50)$$

Thus $a_j = [\delta^j] \mathbb{E} e^Q$, where the formal Gaussian moments are

$$\mathbb{E} X^{2m} = (2m-1)!! \left(\frac{2}{3c} \right)^m, \quad \mathbb{E} X^{2m+1} = 0.$$

For complex c these mean analytic Gaussian integration; $\Re c > 0$ ensures convergence. The leading Jacobian and Gaussian integral give $e^K \sqrt{c/(3\pi)} \delta^{5/2}$.

The convergent local series and the logarithmic/binomial expansions give uniform Taylor remainders of any requested power of δ times a polynomial in X . After multiplication by the Gaussian they are dominated by Ce^{-bX^2} times a polynomial. Expand two extra half-orders and use parity; the resulting remainder is $O(\delta^{J+1})$, without a logarithmic loss. Holomorphy and Cauchy estimates give parameter derivatives. Finally $e^{(s-1)z}$ changes K to $K + (s-1)\rho$ and g_2 to $g_2 - (s-1)\rho$, with all estimates uniform for bounded s . $\square$

Put $T_0 = c^2/2 + g_1\sqrt{c}$. The first two coefficients are

$$a_1 = T_0 - \frac{5}{36c} = c^2 \left(\frac{1}{2} + \frac{\rho(a^2-3)}{4a} \right) - \frac{5}{36c}, \quad (51)$$

$$a_2 = \frac{T_0^2}{2} + cg_2 - \frac{17g_1}{36\sqrt{c}} - \frac{17c}{72} - \frac{35}{2592c^2}. \quad (52)$$

Formula (50) is a finite algorithm at every fixed order; the supplied program also evaluates the third order.

13 The marking phase gap

Lemma 13.1. *For any compact positive interval $\mathcal{U}$ and $\epsilon > 0$, there are $\eta > 0$ and $C < \infty$ such that, uniformly for $u \in \mathcal{U}$ and $\epsilon \leq |\theta| \leq \pi$,*

$$|[z^n]H(z, ue^{i\theta}, s)| \leq C(\rho(u) + \eta)^{-n}, \quad (53)$$

with s in any prescribed compact set.

Proof. For $|z| \leq \rho(u)$, inequality (47) is strict except at $x = 1/2, z = \rho(u)$. At that exceptional point multiplication by $e^{i\theta}$ gives $e^{i\theta} \neq 1$. The denominator is therefore nonzero on the entire compact parameter set, and remains nonzero for $|z| \leq \rho(u) + \eta$ with uniform $\eta > 0$. The integral and exponential are uniformly bounded there. Cauchy's estimate proves the result. $\square$

This excludes other periodic saddles in the marking variable. It is a separate ingredient beyond a positive-real asymptotic formula.

14 Precise local large deviations

Use $v = \log u$ and put

$$f(v) = -\log \rho(e^v), \quad d(v) = \log D(e^v), \quad \alpha(u) = \frac{1}{(1 + \sqrt{u})\rho(u)}, \quad V(u) = f''(v).$$

In this and the next section, primes on c or ρ mean derivatives of $c(e^v)$ or $\rho(e^v)$ with respect to v , evaluated at $v = \log u$. Then $f' = \alpha$. With $a = u^{-1/2}$ and $\ell = \log(1 + a)$,

$$V(u) = \frac{a(a - \ell)}{4(1 + a)^2 \ell^2} > 0. \quad (54)$$

The map $\alpha : (0, \infty) \rightarrow (0, 1/2)$ is increasing and onto.

Theorem 14.1. *Suppose k/n lies in a fixed compact subinterval of $(0, 1/2)$, and choose the unique $u > 0$ with $\alpha(u) = k/n$. For each fixed $J \geq 0$,*

$$\begin{aligned} A_{n,k} := \sum_j A_{n,k,j} &= \frac{n! D(u) \rho(u)^{-n} u^{-k} e^{3c(u)n^{1/3}}}{\sqrt{2\pi V(u)} n^{4/3}} \\ &\times \left(\sum_{j=0}^J b_j(u) n^{-j/3} + O(n^{-(J+1)/3}) \right), \end{aligned} \quad (55)$$

uniformly in these integer pairs. The coefficients are analytic and explicitly computable. In particular

$$b_0 = 1, \quad b_1 = a_1 - \frac{9(c')^2}{2V}, \quad (56)$$

$$b_2 = a_2 - \frac{9a_1(c')^2}{2V} + \frac{81(c')^4}{8V^2} - \frac{3c''}{2V} - \frac{3c'd'}{V} + \frac{3c'f'''}{2V^2}. \quad (57)$$

Proof. Fourier inversion on the u -circle integrates $A_n(ue^{i\theta}, 1)u^{-k}e^{-ik\theta}$. For small θ apply Theorem 12.1. Positivity of V gives

$$\Re(f(v + i\theta) - f(v) - i\alpha\theta) \leq -b\theta^2.$$

Since c, d and their derivatives are real on the real axis, their real-part increments are $O(\theta^2)$. The normalized integrand is bounded by $Ce^{-b\theta^2}$, with a reduced $b > 0$ if necessary. Outside a fixed neighborhood, Lemma 13.1 supplies an exponential-in- n gain that beats every stretched factor. Only $|\theta| \leq n^{-1/2} \log n$ contributes to algebraic orders.

Set $\delta = n^{-1/3}$ and $\theta = \delta^{3/2}X$. After removal of $e^{-VX^2/2}$ the formal exponent is

$$\begin{aligned} P &= \sum_{m \geq 3} \frac{f^{(m)}(v)}{m!} (iX)^m \delta^{3m/2-3} \\ &+ 3 \sum_{m \geq 1} \frac{c^{(m)}(v)}{m!} (iX)^m \delta^{3m/2-1} + \sum_{m \geq 1} \frac{d^{(m)}(v)}{m!} (iX)^m \delta^{3m/2}. \end{aligned} \quad (58)$$

Therefore

$$b_j = [\delta^j] \mathbb{E} \left[e^P \sum_{r \geq 0} a_r(v + i\delta^{3/2}X) \delta^r \right], \quad X \sim N(0, 1/V), \quad (59)$$

where $a_r(v + i\theta)$ means $a_r(e^{v+i\theta})$ and $a_0 = 1$. Only finitely many terms are needed at each order. Odd half-powers have odd X parity and integrate to zero. Uniform analytic Taylor bounds and a Gaussian majorant prove the remainder as in Theorem 12.1. The new Gaussian factor is $n^{-1/2}/\sqrt{2\pi V}$, giving (55); expansion gives (56)–(57). $\square$

The stretched exponential shifts the full marking saddle by $-3c'/(Vn^{2/3}) + O(n^{-1})$ in v . Its first logarithmic correction is consequently $-9(c')^2 n^{-1/3}/(2V)$, agreeing with (56). Keeping the first-saddle coefficients unchanged would miss this term.

For a uniform object, write K_n for its downset count. Dividing by (46) at $u = 1$ gives

$$\mathbb{P}(K_n = k) \sim \frac{D(u)}{D(1)\sqrt{2\pi nV(u)}} \exp\left(-nI(\alpha) + 3[c(u) - c(1)]n^{1/3}\right), \quad (60)$$

$$I(\alpha) = \alpha \log u + \log \rho(u) - \log \rho(1), \quad \alpha = k/n.$$

The rate is nonnegative, vanishes only at $\alpha(1)$, and has $I' = \log u$, $I'' = 1/V(u) > 0$. All relative corrections follow by series division.

15 Gaussian and isolated element laws

Put $\rho_0 = \log 4$ and $c_0 = c(1)$. Uniform analyticity gives

$$\begin{aligned} \mathbb{E}K_n &= \frac{n}{2\rho_0} + \frac{c_0(1 - \rho_0)}{2\rho_0}n^{1/3} + \frac{1}{6} + \frac{1}{12\rho_0} + O(n^{-1/3}), \\ \text{Var } K_n &= V(1)n + 3c''|_{u=1}n^{1/3} + d''(0) + O(n^{-1/3}), \\ V(1) &= \frac{1}{4\rho_0^2} - \frac{1}{8\rho_0} > 0. \end{aligned} \quad (61)$$

Useful derivative identities are $3c' = c(\alpha - 1/2)$ and $3c'' = c[V + (\alpha - 1/2)^2/3]$. The central and lattice local limits are

$$\frac{K_n - n/(2\rho_0)}{\sqrt{nV(1)}} \implies N(0, 1), \quad (62)$$

$$\sup_{k \in \mathbb{Z}} \left| \sqrt{nV(1)} \mathbb{P}(K_n = k) - \phi\left(\frac{k - n/(2\rho_0)}{\sqrt{nV(1)}}\right) \right| \longrightarrow 0. \quad (63)$$

Here $\phi(x) = e^{-x^2/2}/\sqrt{2\pi}$. For (63), the rescaled characteristic functions converge on each compact interval. Their tails vanish in L^1 by the Gaussian bound from Theorem 14.1 and Lemma 13.1. Fourier inversion bounds the uniform local error by this L^1 error. Centering by the true mean gives the same statements.

Let J_n count isolated elements. Equation (37) and the complex-uniform theorem imply

$$\left(\frac{K_n - n/(2\rho_0)}{\sqrt{nV(1)}}, J_n\right) \implies (Z, P), \quad Z \sim N(0, 1), \quad P \sim \text{Poisson}(\rho_0), \quad Z \perp P. \quad (64)$$

The joint transform factors in the limit into $e^{-t^2/2}e^{(s-1)\rho_0}$. Taking real t and s on the unit circle proves the assertion by the characteristic-function criterion.

There is also a uniform conditional law at atypical density. With u determined by $k/n = \alpha(u)$,

$$\mathcal{L}(J_n \mid K_n = k) = \text{Poisson}(\rho(u)) + O_{\text{TV}}(n^{-2/3}). \quad (65)$$

Indeed the analytic multiplier changes the first-saddle data to

$$a_1(u, s) = a_1(u), \quad a_2(u, s) = a_2(u) - (s-1)\rho c, \quad d_s = d + (s-1)\rho.$$

Applying the second saddle and dividing gives, uniformly for bounded s ,

$$\mathbb{E}[s^{J_n} \mid K_n = k] = e^{(s-1)\rho} \left\{ 1 - (s-1) \left(\rho c + \frac{3c'\rho'}{V} \right) n^{-2/3} + O(n^{-1}) \right\}. \quad (66)$$

For each fixed $R > 1$ this error is holomorphic and uniform on $|s| \leq R$; Cauchy coefficient bounds and summation of R^{-j} prove (65). Unconditionally,

$$\mathbb{E}[s^{J_n}] = e^{(s-1)\rho_0} \{ 1 - (s-1)\rho_0 c_0 n^{-2/3} + O(n^{-1}) \}, \quad (67)$$

so the same total-variation rate applies. Finite covariance is not zero despite the limiting independence: differentiating the tilted version gives

$$\text{Cov}(K_n, J_n) = -\frac{1}{2} + O(n^{-2/3}).$$

16 Checks, attribution, and scope

The reproducibility package contains exact refined rows through $n = 400$, obtained from (39)–(40). Independent exhaustive generation of the literal labelled binary words agrees jointly in (k, j) through $n = 8$. A separate implementation of the record-to-cycle map checks every round trip through $n = 8$, including deliberately rotated and reordered cycle inputs. One-cycle counts agree with (39).

The symbolic first-saddle coefficients specialize at $u = 1$ to those of Part I (Theorem 1.1 and Appendix A). Both saddle generators are finite-order algorithms, with third-order output supplied. A separate hand-expanded Gaussian audit checks b_1, b_2 without using the generator. Numerical tests cover complex u , conditional isolated means, and densities 0.2, 0.3, 0.4. For example, at $n = 400, k = 160$, the ratio to the leading factor in (55) is approximately 0.8868044222; the absolute error in that normalized ratio after three corrections is 0.0003942506. At density 0.2, convergence is much slower and the normalized third-order error at $n = 400$ is approximately -0.378 . These calculations check the formulas; they do not certify asymptotic error constants or replace the proofs.

[Write note, 2 October 2026, batch 77P5.] The exact refined rows ($A_{n,k}$ for $n \leq 400$, `exact_rows.json`, 18,071,638 bytes) are *not* shipped: they are regenerated by `code/05-refined-validate_refinement.py`, and at the write the regenerated rows for $n \leq 120$ (and at intake for $n \leq 200$) were equal to the delivered ones; the delivered copy survives in the arrival commit, and the report README gives the commands. The other programs and records named in this section are shipped as `code/05-refined-*.py` and `data/05-refined-*.json`; the coefficient program of the third order is `code/05-refined-refined_coefficients.py`.

The model and its statistics are BCK's. General Gaussian and Poisson consequences are standard once the marked coefficient theorem is available. The substantive extension consists of the direct record/cycle explanation of the transform, complex-uniform coefficient analysis, global marking

phase gap, and the all-orders two-stage local-count theorem with its stretched correction. The conditional correction supplies additional structural information.

The results hold on compact positive u -intervals, equivalently compact interior block densities. They do not cover $k = o(n)$ or $n/2 - k = o(n)$, endpoint transitions, or the other complex z -singularity sectors. No convergence of the correction series or exponentially complete expansion is asserted. The original release remains separate and unchanged.

[Write note, 2 October 2026, batch 77P5.] In this report the original release is Part I, printed with the editorial changes listed in “About this report” and no change to any statement.

[Write note, 5 October 2026, batch 101.] At $u = 1$ the other complex z -singularity sectors are now treated in Part IV; for $u \neq 1$ no source treats them (Section 34).

Source and overlap screen. BCK’s Proposition 11 marks different catalytic permutation statistics, not the statistic used here. Its earlier bivariate formulas concern the broader class of 3-free naturally labelled posets. OEIS A113226 and the earlier Elizalde bounds were rechecked [1, 3]. The known cumulant identification is recorded through A136127 and the poly-Bernoulli-relative antidiagonals [4, 6].

Focused public searches found no matching refined local-count theorem. ProveIt’s default-branch code searches for A113226, 12-34, and the BCK author names returned no hits on 1 October 2026; a naturally-labelled-poset search returned unrelated material. The connector reported head 63b9d68407f2 [8]. This is a keyword/index screen, not an exhaustive novelty claim or an audit of every archive attachment. Independent mathematical checks in the package are not a substitute for conventional peer review.

[Write note, 2 October 2026, batch 77P5.] As for Part I, the search ran against the pin 63b9d6840, kept here as provenance; the repository now contains this report and the sibling report `oeis-sequence-asymptotics/a113227-powered-catalan/`.

Part III

A diagonal-state route, block integrals and non-D-finiteness (Report 87)

[Write note, 5 October 2026, batch 101.] **Provenance.** This Part prints, in full, Report 87 of the session bundle of batch 101, *Exact Enumeration and Complete Asymptotics for Vincular Pattern Avoidance*, dated 1 October 2026 (archive `Exact_Vincular_Avoidance_Asymptotics_Source.zip`, 558,023 bytes, arrival commit 60f54ea06, placed in f7c612c72; its delivered PDF has 8 pages). Its author line reads “Research note prepared for Vladimir Reshetnikov with OpenAI”; it carries no “prepared for private review” line. Its repository pin is ProveIt 1512ef835 (1 October 2026, 08:38 PDT, recorded in its source-status note and in the bibliography entry [12]); that commit is an ancestor of Part I’s pin 63b9d6840 (15:32 PDT the same day). The archive was delivered on 5 October 2026, after Parts I–II had been written.

Relation to Parts I and II. Report 87 and source 04 (Part I) are independent, same-day answers to the same assignment: neither cites the other, and at Report 87’s pin the repository contained no A113226 material (its claim that the ProveIt inventory has “no named

A113226/A136127 target” is true at that pin and stale now, since this report was placed in 4f11bc9c0). Most of its results are therefore *second routes* to Part I, printed here because their proofs differ; no novelty over Part I is claimed for them. By pin, Report 87 is the earlier of the two; by printing order it comes second. In detail:

- Theorem 17.1 is Theorem 1.1, (2), by a *new route*: continuous labels with the state (lowest ascent top, last label) and its separately retained diagonal (Section 18), directly from the permutation definition, where Part I uses the labelled binary-word model of [2, Proposition 7]. Under $t = e^{z/2} = 2q$ the closed form (69) is (2).
- (79), (80) and (81) are (3), (12) and (13) up to an index shift ($t_n = \ell_n$).
- (83) is (14) by a second literature route (the Clark–Ehrenborg bivariate formula, where Part I uses Testart’s antidiagonal identity).
- Theorem 17.2 and (72) are (4) and (5), with a different contour (left arc plus vertical chord). The delivered `data/06-diagonal-saddle_checks.json` gives c_2 and c_3 exactly; they are symbolically identical to Part I’s (6) and a_3 of Appendix A.
- Section 20’s continuation through the circle $|z| = \rho$ re-proves Section 4.
- Theorem 23.1 and (90) are variants of Section 7 (a gamma-core reversion instead of the Lambert- W core), and like it instances of `p0:thm:staircase` and `p0:thm:perturbed-inversion` of the transseries volume (see the write note at the end of Section 7).

New in this Part: the permutation-level diagonal-state derivation (Section 18); the integral forms of Elizalde’s block functions and the exact identity (82) (Section 19); and Proposition 22.1: the EGF is differentially algebraic but not D-finite, so A113226 is not P-recursive. Parts I–II state nothing of the kind; Part I only notes that the ODE continues L along paths avoiding $\rho + 2\pi ik$.

Numbering and editorial changes. Report 87’s Section n is Section $n + 16$, its equation (k) is equation ($k + 67$), and its Theorems 1.1, 1.2, Proposition 6.1 and Theorem 7.1 are Theorems 17.1, 17.2, Proposition 22.1 and Theorem 23.1. Its 28 labels carry the prefix `vav:dg::`; its citations of OEIS A113226, Bevan–Cheon–Kitaev, Elizalde and A136127 point to Part I’s entries for the same works, and its entries for Clark–Ehrenborg, Ferraz de Andrade–Lundberg–Nagle, Flajolet–Sedgewick and its ProveIt pin are added as [9], [10], [11] and [12] (the source lists the first, third and fourth without citing them). Its macros `\ee`, `\ii`, `\E` are kept (`\ii` and `\E` are Part I’s). No statement is changed. Write notes (dated) and three numbered write results (Remarks 19.1 and 20.1, Proposition 23.2, each marked [write], placed after the source’s numbered items so that no source number moves) are added. Unproved items go to Section 24.1.

Shipped records. The working derivations are `06-diagonal-exact_egf_proof.md` and `06-diagonal-all_orders_proof.md` (they write a for κ), the structural corollaries `06-diagonal-structural_corollaries.md`, the source screen `06-diagonal-source_status.md`, and two internal reviews `06-diagonal-audit-structural_exact_review.md` and `06-diagonal-audit-structural_all_orders_review.md`; the programs are `code/06-diagonal-check_*.py` and their outputs `data/06-diagonal-*.json`. The reviews are AI-internal; the Part is unrefereed and not formalized.

[Write note, 5 October 2026, batch 101.] **Reading conventions of Part III.** Report 87 keeps its own letters. Table 2 maps them to Parts I–II and prints the tempting false readings. Three clashes deserve emphasis: Report 87’s C is Part I’s amplitude D , while Part I’s C is $3c$; Report 87’s a_n are the counts (Part I’s A_n), while Part I’s a_j are the correction coefficients (Report 87’s c_j); and inside Report 87, $b_k(z)$, $c_k(z)$ are Elizalde’s block functions while b_n counts A136127 and c_j are the correction coefficients.

Table 2: Reading conventions of Part III (Report 87) against Parts I–II.

Part III	meaning in Part III	in Parts I–II (false reading)
$a_n, A(z)$	counts of avoiders, their EGF	$A_n, H(z)$; Part I's a_j are coefficients, Part II's $a = u^{-1/2}$; Part I's $A = \pi/\sqrt{\rho}$
$T(z), t_n, T_0$	$\log A$, its coefficients $T^{(n)}(0)$, $\rho/2 - 3$	$L(z), \ell_n, K$; Part I's $T(x, y)$ is the bivariate array EGF
κ, c_j	$(\pi^2/(4\rho))^{1/3}$, correction coefficients	c, a_j ; Part III's c in (86) is a generic constant
C, C_0, D_K	amplitude $e^{\rho/2-3}\sqrt{\kappa/(3\pi)}$; generic constants	C is Part I's D (not Part I's $C = 3c$); C is also a generic constant in Section 21
$D(m)$	$e^{z(m-1)} + e^{-zm} - 1$	$e^{-z}D_z(m)$ of Section 2; not the amplitude D
$F(m, y), G(m), H(m)$	continuation series and diagonal	not Part I's smooth realization F or EGF $H(z)$
w, δ	$\sqrt{4e^{-z} - 1}$; $\rho - z$ (Section 20)	Part I's $w = \rho - z$ and $\delta = n^{-1/3}$
h	$n^{-1/6}$	the same as Part I's $h = \sqrt{\delta}$ in Section 6
u	A'/A (Proposition 22.1); $u(h, v)$ (Section 21); integration variable	not Part II's marking parameter u
$b_k, c_k / b_n, B$	Elizalde's block functions / A136127 counts and their EGF	$b_n = \ell_{n+1}$; Part I's $b(x)$; Part IV's $b_n = a_n/n!$
$E, P_E, \mathbb{E}$	$e^z, x^2 - Ex + E$; expectation	$\mathbb{E}$ is Part I's expectation
$Q, M, R(h, v)$	excedance EGF, its primitive; the formal exponent	Part I's $Q(\delta, U)$, blocks $M_i, R(w)$
$N(y), X, q(X), \Psi$	threshold $\min\{n : a_n \geq y\}$; gamma-core inverse; its perturbation; $\psi(X+1) - \log \rho$	Part I's $\tau(Y)$; Part I's N (Lambert core), X (an approximation), $q = e^{z/2}/2$
Z, γ	standard normal; $\sqrt{2\kappa/3}$	Part II's $Z \sim N(0, 1)$ in (64)

Abstract of Report 87. Let a_n count permutations avoiding the vincular pattern 12–34, the sequence OEIS A113226. A continuation argument with independent uniform labels gives an exact exponential generating function

$$\sum_{n \geq 0} a_n \frac{z^n}{n!} = \exp \left(\int_1^{e^z} \frac{x \, dx}{x^2 - e^z x + e^z} \right).$$

The diagonal state in which the last entry is the lowest ascent top is essential to the argument. The formula resolves the growth and stretched-exponential conjectures stated by Bevan, Cheon and Kitaev in 2025, with explicit constants and a complete expansion in powers of $n^{-1/3}$. We give a finite Gaussian-moment algorithm, an exact recurrence, a connection with the initial-excedance-set numbers, and inverse expansions with rigorous integer-threshold bounds. The generating function is differentially algebraic but not D-finite, so the counting sequence is not P-recursive. The singularity and contour estimates are proved directly; no conjectured recurrence is assumed.

17 Statements and source context

A permutation $\pi \in S_n$ contains 12–34 if there are indices $i+1 < j$ such that $\pi_i < \pi_{i+1} < \pi_j < \pi_{j+1}$. Write a_n for the number avoiding this pattern and $A(z) = \sum_{n \geq 0} a_n z^n / n!$. The initial values are

$$1, 1, 2, 6, 23, 107, 585, 3669, 25932, 203768, 1761109.$$

This is A113226 [1]; it also counts $\{3, 2+2\}$ -free naturally labelled posets [2]. Throughout, logarithms are natural.

Theorem 17.1 (Exact generating function). *As an analytic germ at zero,*

$$A(z) = e^{T(z)}, \quad T(z) = \int_1^{e^z} \frac{x \, dx}{x^2 - e^z x + e^z}. \quad (68)$$

Equivalently, with the branches specified by $w > 0$ for $0 < z < \rho = \log 4$,

$$T(z) = \frac{z}{2} + \frac{\pi - 3 \arctan w}{w}, \quad w = \sqrt{4e^{-z} - 1}. \quad (69)$$

Set

$$\rho = \log 4, \quad \kappa = \left(\frac{\pi^2}{4\rho} \right)^{1/3}, \quad C = e^{\rho/2-3} \sqrt{\frac{\kappa}{3\pi}}. \quad (70)$$

Theorem 17.2 (Complete expansion). *There are explicitly computable real constants c_j , with $c_0 = 1$, such that for every fixed nonnegative integer K ,*

$$\frac{a_n}{n!} = C \rho^{-n} e^{3\kappa n^{1/3}} n^{-5/6} \left(\sum_{j=0}^K c_j n^{-j/3} + O_K(n^{-(K+1)/3}) \right). \quad (71)$$

The first correction is

$$c_1 = \frac{\kappa^2(1 - \rho)}{2} - \frac{5}{36\kappa} = -0.398274243294777\dots \quad (72)$$

Section 21 gives a finite exact algorithm for every c_j .

In the form $a_n \sim C \rho^{-n} \mu^{n^{1/3}} n^\beta n!$, the constants are

$$\mu = e^{3\kappa} = 37.926693850215157\dots, \quad \beta = -\frac{5}{6}, \quad C = 0.0357060412580448\dots$$

Bevan–Cheon–Kitaev [2, Conjectures 13–14] conjecture the growth constant ρ^{-1} and this form, with numerical estimates $\mu \approx 37.9$, $\beta \approx -0.82$ and amplitude near 0.032. Equation (71) supplies exact constants; the finite-data fitted amplitude is not identified with the limiting one.

Elizalde [3, Proposition 5.1] supplies the earlier block decomposition and coefficientwise envelopes. The insertion state used below is the same lowest-ascent-top/last-entry state appearing in [2, Proposition 11]; the present argument replaces its discrete catalytic variables by continuous labels and solves the resulting integral equations. The saddle method and the reversion mechanism are classical. The proposed contribution is the exact generating function for this class, its complete explicit expansion and the controlled inverse application. Bounded searches of the live OEIS entry, its primary references and later citations on 1 October 2026 did not locate this formula or a later proof of the stated conjectures. This is a source-status report, not a claim of exhaustive priority.

[Write note, 5 October 2026, batch 101.] In this report the exact generating function, the complete expansion and the controlled inverse are Part I’s Theorem 1.1 and Section 7, from a manuscript of the same day; the proposed contribution of this paragraph is therefore a second, independent proof of those results (see the note at the head of this Part). The constants agree: $\kappa = c$, $C = D$, $c_1 = a_1$, and $\mu = 37.926693850215157\dots$ as in Section 1.

18 Continuous labels and the diagonal state

Independent uniform labels in $(0, 1)$ have each order type of length n with probability $1/n!$. Thus the probability of avoidance is $a_n/n!$. After an allowed nonempty prefix, let y be its last label and m the minimum of the ascent tops already seen. Put $m = 1$ if no ascent has occurred. A proposed new ascent starting at y is allowed exactly when $y \leq m$. Equality is allowed: the ascent ending at $y = m$ and the next ascent overlap, so cannot alone form a four-entry occurrence.

Let $F(m, y; z)$ count allowed continuations, including the empty continuation, and retain the diagonal value $G(m; z) = F(m, m; z)$ separately. The diagonal is attained by a positive proportion of prefixes, even though it has two-dimensional measure zero. For $|z| < 1$ all continuation series are uniformly absolutely convergent, bounded by $(1 - |z|)^{-1}$. We suppress z in the notation. Appending a uniform label gives, for $0 < y < m$,

$$F(m, y) = 1 + z \int_0^y F(m, u) du + z \int_y^m G(u) du + z \int_m^1 F(m, u) du. \quad (73)$$

A descent keeps m ; an ascent to $u < m$ creates the diagonal state (u, u) ; an ascent to $u > m$ keeps m . For $m < y < 1$ only a descent is permitted:

$$F(m, y) = 1 + z \int_0^y F(m, u) du. \quad (74)$$

On the diagonal every append is permitted, and apart from null equalities the threshold stays m :

$$G(m) = 1 + z \int_0^1 F(m, u) du. \quad (75)$$

These equations uniquely specify the continuation series coefficient by coefficient, so no extraneous solution can enter.

Put $H(m) = 1 + z \int_0^m F(m, u) du$. Equation (74) and then (75) give

$$F(m, y) = H(m)e^{z(y-m)} \quad (y > m), \quad G(m) = H(m)e^{z(1-m)}. \quad (76)$$

The left limit of (73) at $y = m$ is $G(m)$, and differentiation in y gives $\partial_y F = z(F - G(y))$. Hence

$$F(m, y) = e^{z(y-m)}G(m) + ze^{zy} \int_y^m e^{-zu}G(u) du \quad (y < m). \quad (77)$$

Integrating this identity yields

$$H(m) = 1 + (1 - e^{-zm})G(m) + z \int_0^m (1 - e^{-zu})G(u) du.$$

Together with (76), this becomes the scalar Volterra equation

$$D(m)G(m) = 1 + z \int_0^m (1 - e^{-zu})G(u) du, \quad D(m) = e^{z(m-1)} + e^{-zm} - 1. \quad (78)$$

Near $z = 0$, D is uniformly nonzero. Thus $G(0) = e^z$ and

$$\frac{G'(m)}{G(m)} = \frac{z(1 - e^{z(m-1)})}{D(m)}.$$

Set $E = e^z$, $x = e^{zm}$ and $P_E(x) = x^2 - Ex + E$. Then $d \log G = (E - x)dx/P_E(x)$, so

$$\log G(1) = z + \int_1^E \frac{E - x}{P_E(x)} dx.$$

Since $P_E(E) = E$, $P_E(1) = 1$ and $\int_1^E (2x - E)dx/P_E(x) = z$, the last expression equals $T(z)$ in (68). Finally, starting with the first uniform label,

$$A(z) = 1 + z \int_0^1 F(1, y) dy = H(1) = G(1).$$

This proves (68) as a germ at zero. Later continuation is deduced from that identity, rather than from the original continuation series outside $|z| < 1$.

[Write note, 5 October 2026, batch 101.] This derivation does not use Bevan–Cheon–Kitaev’s word bijection of [2, Proposition 7], on which Part I’s Section 2 rests; its state is the one of their insertion tree [2, Proposition 11], as the source says. Its Volterra denominator is Part I’s: $D(m) = e^{-z} D_z(m)$ with D_z as in Section 2, and the substitution $x = e^u$ turns (68) into $T(z) = \int_0^z e^u du / (e^u + e^{z-u} - e^z)$, which is (9) after symmetrizing $u \leftrightarrow z - u$ (both equal $z \int_0^1 e^{zx} dx / D_z(x)$; compare (85)). The sufficiency of the state (m, y) , including the admissibility of $y = m$, is checked in the shipped review 06-diagonal-audit-structural_exact_review.md; it was rechecked at intake.

19 Exact recurrences and the published block counts

Elementary differentiation of (69), or of (68), gives

$$(4 - e^z)T'(z) - 2T(z) = 2 + e^z - z, \quad T(0) = 0. \quad (79)$$

Writing $T(z) = \sum t_n z^n / n!$, we have $t_0 = 0$, $t_1 = 1$, and

$$3t_{n+1} = \sum_{j=0}^{n-1} \binom{n}{j} t_{j+1} + 2t_n + 1 - \mathbf{1}_{\{n=1\}} \quad (n \geq 1). \quad (80)$$

The desired counts follow from

$$a_0 = 1, \quad a_n = \sum_{j=1}^n \binom{n-1}{j-1} t_j a_{n-j}. \quad (81)$$

Both recurrences are exact. In particular, they do not constitute a recurrence guessed from numerical data.

For comparison, Elizalde’s functions b_k, c_k satisfy $b_0 = z$, $c_0 = e^z - 1 - z$ and, for $k \geq 1$,

$$b_k'' = k^2 e^z b_{k-1}, \quad c_k'' = k(k+1) e^z c_{k-1},$$

with zero values and first derivatives at zero. They admit the integrals

$$b_k = \int_0^z (e^u - 1)^k (e^{z-u} - 1)^k du, \quad c_k = \int_0^z (e^u - 1)^{k+1} (e^{z-u} - 1)^k du.$$

Checking the recurrences and initial conditions proves these identities. Geometric summation near zero then gives

$$\sum_{k \geq 0} (b_k + c_k) = \int_0^z \frac{e^u du}{1 - (e^u - 1)(e^{z-u} - 1)} = T(z). \quad (82)$$

Thus, if $S = \sum_{k \geq 1} (b_k + c_k)$, the exact formula is $A = \exp(S + e^z - 1)$. It lies between Elizalde’s envelopes $\exp S$ and $\exp(S + e^z + z - 1)$, but is not inferred from them.

19.1 Initial excedance sets

Let b_n count permutations of $[n]$ whose excedance set is an initial interval, with $b_0 = 1$; this is A136127 [4]. The classical extremal-excedance generating function, due to Clark–Ehrenborg and reproduced in [10, Eq. (3.1)], is

$$Q(x, y) = \sum_{r, s \geq 0} [b^r a^s] \frac{x^r y^s}{r! s!} = \frac{e^{-x-y}}{(e^{-x} + e^{-y} - 1)^2}.$$

Here $[b^r a^s]$ counts permutations of $r + s + 1$ with precisely the first r positions excedances. A beta integral shows

$$B(z) := \sum_{n \geq 0} b_n \frac{z^n}{n!} = 1 + \int_0^z Q(u, z - u) du.$$

Put $M(x, y) = e^{-y}/(e^{-x} + e^{-y} - 1)$. Equation (82) reads $T(z) = \int_0^z M(z - v, v) dv$; since $M(0, z) = 1$ and $\partial_x M = Q$, we get

$$T'(z) = B(z), \quad t_n = b_{n-1} \ (n \geq 1), \quad A' = BA. \quad (83)$$

The bivariate excedance formula is established prior work. The connection (83) with this single-pattern avoidance class follows from the exact continuation solution above.

Remark 19.1 ([write] The block integrals and the block grading). (a) *Proof of the integral forms.* The source states that checking the recurrences proves them; here is the check. For integers $m, n \geq 0$ put $a = e^u - 1$, $b = e^{z-u} - 1$ and $I_{m,n}(z) = \int_0^z a^m b^n du$, so that the displayed integrals are $b_k = I_{k,k}$ and $c_k = I_{k+1,k}$, and $I_{0,0} = z = b_0$, $I_{1,0} = e^z - 1 - z = c_0$. We claim that for $m, n \geq 1$

$$I''_{m,n} = mn e^z I_{m-1,n-1}, \quad I_{m,n}(0) = I'_{m,n}(0) = 0.$$

The initial values are clear. Since $\partial_z a = 0$ and $\partial_z b = 1 + b$, and $b = 0$ at $u = z$,

$$\begin{aligned} I'_{m,n} &= \int_0^z n a^m b^{n-1} (1 + b) du, \\ I''_{m,n} &= \mathbf{1}_{\{n=1\}} (e^z - 1)^m + \int_0^z [n(n-1) a^m b^{n-2} (1 + b)^2 + n a^m b^{n-1} (1 + b)] du, \end{aligned}$$

where the boundary term comes from $u = z$ and survives only for $n = 1$. On the other hand $\partial_u a = 1 + a$ and $\partial_u b = -(1 + b)$ give

$$n \partial_u [a^m b^{n-1} (1 + b)] = mn(1 + a)(1 + b) a^{m-1} b^{n-1} - n(n-1) a^m b^{n-2} (1 + b)^2 - n a^m b^{n-1} (1 + b).$$

Integrate over $[0, z]$. The bracket vanishes at $u = 0$ (where $a = 0$) and at $u = z$ (where $b = 0$) unless $n = 1$, when it equals $(e^z - 1)^m$ there. Since $(1 + a)(1 + b) = e^z$, the integral of the first term on the right is $mn e^z I_{m-1,n-1}$, and the boundary contribution cancels the term $\mathbf{1}_{\{n=1\}} (e^z - 1)^m$ above. This proves the claim. With $(m, n) = (k, k)$ and $(k + 1, k)$ it is Elizalde's pair of recurrences, and the solution of $y'' = f$ with $y(0) = y'(0) = 0$ is unique, so induction on k proves the integral forms. (The shipped proof note 06-diagonal-exact_egf_proof.md also mentions a combinatorial reading by a distinguished threshold label, without detail.) Summing, $\sum_k (b_k + c_k) = \int_0^z (1 + a) \sum_k (ab)^k du = \int_0^z e^u du / (1 - ab)$ for small $|z|$, and the substitution $x = e^u$ in (68) gives the same integral, since $1 - ab = e^u + e^{z-u} - e^z$ and $x^2 - e^z x + e^z = e^u (e^u + e^{z-u} - e^z)$. This is (82).

(b) *The index k is not Part II's block-pair grading.* One might read $\sum_k u^k (b_k + c_k)$ as $\log H(z, u)$ of (38). It is not, except at $u = 1$: since $b_k, c_k = O(z^3)$ for $k \geq 1$, $\sum_k u^k (b_k + c_k) = e^z - 1 + O(z^3)$

has z^2 -coefficient $1/2$, whereas $\log H(z, u) = z + L_2(u)z^2/2 + O(z^3)$ with $L_2(u) = u$ by (39). (At intake the two were also compared numerically at $u = 0.7$; they differ there, and agree at $u = 1$.) In fact the grading of Part II is a shifted one: substituting $s = zx$ in (38) and using $e^z - e^s = (1 + a)b$ (with $a = e^s - 1$, $b = e^{z-s} - 1$) gives, for small $|z|$, $\log H(z, u) = z + \sum_{k \geq 0} u^{k+1}(I_{k,k+1} + I_{k+1,k+1})$, and $I_{k,k+1} = I_{k+1,k}$ by $s \leftrightarrow z - s$; hence

$$\log H(z, u) = b_0 + \sum_{k \geq 1} u^k (b_k + c_{k-1}) :$$

u marks b_k with weight k but c_k with weight $k+1$. (This paragraph was added after the independent check recorded at the end of Section 34.)

[Write note, 5 October 2026, batch 101.] Part I identifies T' (its L') with the EGF of A136127 by Testart's antidiagonal identity [6, Section 6]; the Clark–Ehrenborg route above is a second, independent literature route to (14). The question of a direct bijection explaining $A' = BA$, raised again in Section 24, has the status recorded after Part I's questions in Section 9.

20 Singularity and a complete contour argument

For $0 < z < \rho$, direct integration of (68) gives

$$T(z) = \frac{z}{2} + \frac{E}{\sqrt{E(4-E)}} \left[\arctan \frac{E}{\sqrt{E(4-E)}} - \arctan \frac{2-E}{\sqrt{E(4-E)}} \right].$$

The bracket is $\pi - 3 \arctan w$, proving (69). Writing $\delta = \rho - z$, its convergent local expansion is

$$\begin{aligned} T(\rho - \delta) &= \frac{\rho - \delta}{2} + \frac{\pi}{\sqrt{e^\delta - 1}} - 3 \sum_{j \geq 0} \frac{(-1)^j (e^\delta - 1)^j}{2j + 1} \\ &= \frac{\pi}{\sqrt{\delta}} + \frac{\rho}{2} - 3 - \frac{\pi}{4} \sqrt{\delta} + \frac{\delta}{2} + O(\delta^{3/2}). \end{aligned} \tag{84}$$

The square root is positive for positive real δ .

To control other points of the convergence circle, use

$$T(z) = z \int_0^1 \frac{e^{zv}}{1 - (e^{zv} - 1)(e^{z(1-v)} - 1)} dv. \tag{85}$$

For $|z| = r < \rho$,

$$|(e^{zv} - 1)(e^{z(1-v)} - 1)| \leq (e^{rv} - 1)(e^{r(1-v)} - 1) \leq (e^{r/2} - 1)^2 < 1.$$

On $|z| = \rho$, equality in the second inequality requires $v = 1/2$. There, equality in the first requires the degree-one and degree-two terms of $e^{z/2} - 1$ to have the same argument, forcing $z = \rho$. Thus (85) continues through every other point of the circle. Combining compactness with (84), there is a fixed $\epsilon > 0$ for which A continues to $|z| < \rho + \epsilon$ slit along $[\rho, \rho + \epsilon)$.

Use a Cauchy contour consisting of the left part of the circle $|z| = \rho + \epsilon$ and its vertical chord

$$\Re z = r = \rho(1 - \kappa n^{-2/3}).$$

The slit is excluded. The outer arc stays away from the singularity and contributes an exponentially small error compared with the claimed main term. On the chord $z = r + iy$, the local expansion gives

$$\Re T(r + iy) \leq \frac{\pi}{\sqrt{\rho - r}} - \frac{c}{\sqrt{\rho - r}} \min \left\{ \frac{y^2}{(\rho - r)^2}, 1 \right\} + C_0 \quad (86)$$

for small $|y|$, with fixed $c, C_0 > 0$. This follows from the strict inequality $\Re(1 - it)^{-1/2} < 1$ for $t \neq 0$ and its quadratic expansion at zero. The remaining local terms are bounded. Also $|r + iy| \geq r$. Consequently the part $|y| \geq n^{-5/6+\eta}$, for any sufficiently small fixed $\eta > 0$, loses a factor $\exp(-cn^{2\eta})$ relative to the saddle; farther fixed parts lose $\exp(-cn^{1/3})$. This is the coefficient-contour estimate needed in addition to the real-axis expansion.

Remark 20.1 ([write] The farther parts of the chord). The last claim, which the source states without detail, holds with a much larger loss. Take the arc on a circle $|z| = \rho + \epsilon_1$ with $0 < \epsilon_1 < \epsilon$, so that the closed contour lies in the slit disc of analyticity (on the source's circle $|z| = \rho + \epsilon$ itself analyticity has not been shown; the shrinking is harmless, as the circle radius plays no other role). Fix $y_0 > 0$ so small that (86) holds for $|y| \leq y_0$, and fix $0 < \eta_0 < \rho$. The set $K_1 = \{x + iy : \rho - \eta_0 \leq x \leq \rho, y_0 \leq |y|, |x + iy| \leq \rho + \epsilon_1\}$ is a compact subset of the slit disc (it avoids the real slit since $|y| \geq y_0$), so $|A| \leq M$ on K_1 . For large n the chord points with $|y| \geq y_0$ lie in K_1 , and $|r + iy|^2 \geq r^2 + y_0^2$ there. Since the chord has length at most $2(\rho + \epsilon_1)$ and $r^{-n} = \rho^{-n} \exp(\kappa n^{1/3} + O(n^{-1/3}))$, its part with $|y| \geq y_0$ contributes to $a_n/n!$ at most

$$\frac{M(\rho + \epsilon_1)}{\pi r} \rho^{-n} e^{\kappa n^{1/3} + O(n^{-1/3})} \left(1 + \frac{y_0^2}{(\rho + \epsilon_1)^2}\right)^{-(n+1)/2},$$

which is smaller than the main term $C\rho^{-n}e^{3\kappa n^{1/3}}n^{-5/6}$ by a factor $O(e^{-c_1n})$ with $c_1 = \frac{1}{2} \log(1 + y_0^2/(\rho + \epsilon_1)^2) > 0$; a fortiori it loses $\exp(-cn^{1/3})$. The parts with $n^{-5/6+\eta} \leq |y| \leq y_0$ are those treated by (86).

[Write note, 5 October 2026, batch 101.] This section re-proves Part I's Section 4 with a different contour: Part I uses a full Cauchy circle of radius $\rho(1 - c\delta^2)$ deformed near ρ to a vertical segment, Report 87 the left arc of $|z| = \rho + \epsilon$ and the vertical chord through $\rho(1 - \kappa n^{-2/3})$. The local expansion (84) is (16)–(17) in other variables. The points $\rho + 2\pi ik$, $k \neq 0$, are genuine singularities of every continuation: see Proposition 22.1 and, for the explicit branch, Lemma 26.1 of Part IV.

21 The finite coefficient algorithm

Let $h = n^{-1/6}$, $\gamma = \sqrt{2\kappa/3}$, and define

$$u(h, v) = \kappa h^4 + i\gamma h^5 v, \quad T_0 = \rho/2 - 3.$$

The orientation of the chord can be chosen so that $z = \rho(1 - u)$. Form the formal series

$$\begin{aligned} R(h, v) = & T(\rho(1 - u(h, v))) - (h^{-6} + 1) \log(1 - u(h, v)) \\ & - 3\kappa h^{-2} - T_0 + \frac{v^2}{2}. \end{aligned} \quad (87)$$

The negative powers cancel. The coefficient of h^j in $R \in h\mathbb{C}[v][[h]]$ has the parity of j as a polynomial in v . Let Z be standard normal. Then

$$c_j = \mathbb{E}([h^{2j}] \exp R(h, Z)). \quad (88)$$

This uses only finitely many terms of (84); Gaussian moments are 0 in odd degree and $(2k - 1)!!$ in degree $2k$. It proves (72) by finite algebra.

For completeness, the remainder is controlled as follows. In the central window $|v| \leq h^{-\eta}$, choose $\eta > 0$ sufficiently small for a fixed desired truncation order. Taylor expansion of the convergent local expression and then its exponential gives, through degree J , a remainder bounded by h^{J+1} times a fixed polynomial in $|v|$ times $\exp(Ch|v|^3)$. The full exponent retains Gaussian decay $\exp(-v^2/4)$, either by its local quadratic estimate or by (86). The remainder thus has integral $O(h^{J+1})$. Expand through $J = 2K + 1$. Odd terms integrate to zero on the symmetric window; extending each polynomial-Gaussian integral to the real line costs less than every power of h . The discarded contour is handled by (86). Finally dz supplies $\rho\gamma h^5$, and $1/z$ is already included in (87). The leading multiplier is $\gamma/\sqrt{2\pi} = \sqrt{\kappa/(3\pi)}$. This proves Theorem 17.2, including its remainder at every order.

The result is a dominant-saddle asymptotic expansion. It does not assert convergence of that expansion or describe all exponentially smaller complex saddle contributions.

[Write note, 5 October 2026, batch 101.] The Gaussian variable here is standard (Z , variance 1); Part I's U has variance $2/(3c)$, and $i\gamma h^5 Z$ corresponds to Part I's $c\delta^{5/2}iU$ after $U = \gamma Z/\kappa$. The shipped code/06-diagonal-check_saddle.py runs this algorithm to order 3 (writing a for κ); its output data/06-diagonal-saddle_checks.json lists $c_0, \dots, c_3$, and at the write its c_2 and c_3 were found symbolically equal to Part I's a_2 (6) and a_3 (Appendix A). The “exponentially smaller complex saddle contributions” excluded in the last paragraph are the subject of Part IV.

22 Differential structure and infinitely many singularities

Proposition 22.1. *The germ A is differentially algebraic and is not D -finite. Consequently (a_n) is not P -recursive. Its logarithmic derivative $u = A'/A$ satisfies*

$$3(2u + 1)u'' - 10(u')^2 - (2u + 8)u' + 2u^2 + u - 1 = 0, \quad u(0) = u'(0) = 1. \quad (89)$$

Proof. Differentiating (79) gives $(4 - e^z)u' - (e^z + 2)u = e^z - 1$. Hence near zero,

$$e^z = \frac{4u' - 2u + 1}{u' + u + 1},$$

whose denominator at zero is 3. Differentiate and use $(e^z)' = e^z$ to obtain (89). Substitution $u = A'/A$ and multiplication by A^4 gives a third-order polynomial differential equation for A .

For the other assertion put $z_k = \rho + 2\pi i k$, $k \in \mathbb{Z}$. The square root $w = \sqrt{4e^{-z} - 1}$ can be continued from zero along every finite path avoiding these points. It never reaches i or $-i$, because either value would require $e^{-z} = 0$. The arctangent can therefore be continued along the lifted path as well. Each z_k is accessible by such a path. Near z_k , use w as a local ramified coordinate, so $z = z_k - \log(1 + w^2)$. The continued arctangent is regular at $w = 0$ and has value $j\pi$ for some integer j , by the identity $\tan(\arctan w) = w$. Thus every such branch satisfies

$$T(z) = \frac{z}{2} + \frac{\pi(1 - 3j)}{w} - 3 + O(w^2).$$

Since $1 - 3j \neq 0$, $\exp T$ has an essential singularity in the coordinate w and cannot extend holomorphically through z_k . A D -finite germ can have finite singularities only at the finitely many zeros of the leading polynomial of its linear differential equation. This excludes D -finiteness. Finally, P -recursiveness is preserved by multiplication or division by $n!$, by clearing shifted factorial ratios in the recurrence. P -recursiveness of a_n would therefore make A D -finite. $\square$

[Write note, 5 October 2026, batch 101.] This proposition is new relative to Parts I–II, which state nothing on D-finiteness. The shipped `code/06-diagonal-check_structure.py` verifies the elimination exactly and prints the cleared third-order polynomial equation for A (`data/06-diagonal-structure_checks.json`); at intake the residual of (89) on the exact series was below 10^{-38} through z^{37} . The non-D-finiteness that [2] discuss concerns a different class (the shipped source screen and internal review say so: interval orders, the unlabelled Fishburn series), so it is not a prior proof of this proposition. The local strengths $\pi(1 - 3j)$ are consistent with Part IV: on the branch of Lemma 26.1 the strength is π at $\rho + 2\pi ik$ for even k ($j = 0$) and 2π for odd k ($j = 1$, with the opposite sign of w).

23 Inversion and integer thresholds

For $y \rightarrow \infty$, put $N(y) = \min\{n : a_n \geq y\}$. The expansion implies $a_{n+1}/a_n \sim n/\rho$, so the sequence is eventually strictly increasing. For fixed K , let

$$f_K(x) = \Gamma(x+1)C\rho^{-x}e^{3\kappa x^{1/3}}x^{-5/6}\sum_{j=0}^K c_j x^{-j/3}.$$

It is positive and strictly increasing on a tail, with logarithmic derivative $\log(x/\rho) + O(x^{-2/3})$. Let $x_K(y)$ denote its large inverse.

Theorem 23.1 (Threshold inversion). *For each fixed K , there is a constant $D_K > 0$ such that, for all large y ,*

$$\lceil x_K - \varepsilon_K \rceil \leq N(y) \leq \lceil x_K + \varepsilon_K \rceil, \quad \varepsilon_K = \frac{D_K x_K^{-(K+1)/3}}{\log x_K}.$$

In particular, the ceiling of x_K is exact whenever its distance from the integers exceeds ε_K .

Proof. The logarithmic error at integer arguments in (71) is $O(x^{-(K+1)/3})$. The mean value theorem divides this error by the logarithmic derivative of f_K , which is asymptotic to $\log x$. Taking D_K sufficiently large gives the two comparisons on each side of the threshold. Values in an initial finite segment do not matter as $y \rightarrow \infty$. $\square$

For an explicit compatible inverse expansion, let X solve

$$\log \Gamma(X+1) - X \log \rho = \log y, \quad \Psi(X) = \psi(X+1) - \log \rho,$$

and set

$$q(X) = 3\kappa X^{1/3} - \frac{5}{6} \log X + \log C + \log \left(\sum_{j \geq 0} c_j X^{-j/3} \right).$$

Lagrange reversion in the gamma-core variable gives

$$x \sim X + \sum_{k \geq 1} \frac{(-1)^k}{k!} \left(\frac{1}{\Psi(X)} \frac{d}{dX} \right)^{k-1} \frac{q(X)^k}{\Psi(X)}. \quad (90)$$

Every finite truncation is justified by Taylor expansion of the finite models and the derivative estimate above. The first displacement is $O(X^{1/3}/\log X) = o(X)$, and successive reversion terms decrease. For example,

$$x = X - \frac{3\kappa X^{1/3} - (5/6) \log X + \log C + c_1 X^{-1/3}}{\Psi(X)} + \frac{3\kappa^2 X^{-1/3}}{\Psi(X)^2}$$

$$-\frac{9\kappa^2 X^{2/3} \Psi'(X)}{2\Psi(X)^3} + O\left(\frac{X^{-2/3}}{\log X}\right). \quad (91)$$

No choice of an interpolation of the discrete sequence is being asserted; the finite models and Theorem 23.1 specify the inverse meaning.

Proposition 23.2 ([write] The displayed reversion). *Fix $K \geq 1$, let q_K be q with the sum truncated at $j = K$ (so $\log f_K = G + q_K$ with $G(x) = \log \Gamma(x+1) - x \log \rho$), and let $x = x_K(y)$ and $X = X(y)$ be as above. Then, as $y \rightarrow \infty$,*

- (i) $x = X - \frac{q_K}{\Psi} + \frac{q_K q'_K}{\Psi^2} - \frac{\Psi' q_K^2}{2\Psi^3} + O(X^{-1}(\log X)^{-3})$, all functions evaluated at X ; the two correction groups are the $k = 1$ and $k = 2$ terms of (90) with $q = q_K$;
- (ii) the displayed expansion before this proposition holds for $x = x_K$, with its remainder $O(X^{-2/3}/\log X)$;
- (iii) for every $k \geq 1$ the k -th term of (90), with $q = q_K$, is $O(X^{1-2k/3}(\log X)^{-k})$; so the bounds for successive terms decrease by factors $X^{-2/3}/\log X$, and the bound for the third term is the remainder bound in (i). (The terms themselves need not decrease by this factor: the leading part of the k -th term is

$$\frac{(-1)^k}{k!} (3\kappa)^k \frac{k}{3} \left(\frac{k}{3} - 1\right) \cdots \left(\frac{k}{3} - k + 2\right) X^{1-2k/3} \Psi^{-k}$$

with the $k-1$ factors $\frac{k}{3} - i$, $0 \leq i \leq k-2$; it vanishes when $3 \mid k$, and thus the third term is $9\kappa^3 X^{-1} \Psi^{-4} (1 + O(1/\log X))$, while the fourth is $-\kappa^4 X^{-5/3} \Psi^{-4} (1 + o(1))$.)

Proof. We use $\Psi = G'$, $\Psi(x) = \log(x/\rho) + O(1/x)$, $\Psi'(x) = \psi'(x+1) > 0$ and $\Psi^{(m)}(x) = O(x^{-m})$ for $m \geq 1$ (standard polygamma asymptotics), and $q_K^{(m)}(x) = O(x^{1/3-m})$ for $m \geq 0$; more precisely $q_K = 3\kappa x^{1/3} + O(\log x)$ and $q'_K = \kappa x^{-2/3} + O(x^{-1})$. Write q, q', Ψ, Ψ' for values at X and $d = x - X$.

Size of d . By definition $G(x) + q_K(x) = G(X)$, and $x, X \rightarrow \infty$. If $x \geq 2X$, then $G(x) - G(X) \geq (x-X)\Psi(X) \geq \frac{x}{2}\Psi(X)$, which is incompatible with $G(x) - G(X) = -q_K(x) = O(x^{1/3})$ for large X ; if $x \leq X/2$, then $G(X) - G(x) \geq \frac{X}{2}\Psi(X/2)$, incompatible with $O(x^{1/3}) = O(X^{1/3})$. So $X/2 < x < 2X$, and the mean value theorem with $\Psi \geq \Psi(X/2) \geq \frac{1}{2} \log X$ on $[X/2, 2X]$ gives $d = O(X^{1/3}/\log X)$.

Expansion. Taylor's theorem on $[X/2, 2X]$ gives $G(x) - G(X) = \Psi d + \frac{1}{2} \Psi' d^2 + O(X^{-2}|d|^3)$ and $q_K(x) = q + q'd + O(X^{-5/3}d^2)$, so

$$\Psi d + q + q'd + \frac{1}{2} \Psi' d^2 = O(X^{-1}(\log X)^{-2}).$$

Since $q'd = O(X^{-1/3}/\log X)$ and $\Psi' d^2 = O(X^{-1/3}(\log X)^{-2})$, division by Ψ gives $d = -q/\Psi + d_1$ with $d_1 = O(X^{-1/3}(\log X)^{-2})$. Substituting back, $q'd = -qq'/\Psi + O(X^{-1}(\log X)^{-2})$ and $\Psi' d^2 = \Psi' q^2/\Psi^2 + O(X^{-1}(\log X)^{-3})$, and a second division by Ψ gives (i). The $k = 1$ term of (90) is $-q/\Psi$ and the $k = 2$ term is $\frac{1}{2} \Psi^{-1}(q^2/\Psi)' = qq'/\Psi^2 - \Psi' q^2/(2\Psi^3)$.

(ii). Since $K \geq 1$, $q_K = 3\kappa X^{1/3} - \frac{5}{6} \log X + \log C + c_1 X^{-1/3} + O(X^{-2/3})$, so $-q_K/\Psi$ is the first displayed fraction up to $O(X^{-2/3}/\log X)$. Next $q_K q'_K = 3\kappa^2 X^{-1/3} + O(X^{-2/3} \log X)$ and $q_K^2 = 9\kappa^2 X^{2/3} + O(X^{1/3} \log X)$, with $\Psi' = O(1/X)$; hence the other two groups of (i) equal $3\kappa^2 X^{-1/3}/\Psi^2$ and $-9\kappa^2 X^{2/3} \Psi'/(2\Psi^3)$ up to $O(X^{-2/3}/\log X)$.

(iii). Say that $g \in \mathcal{S}(\mu)$, for a positive weight μ , if $g^{(m)}(X) = O(\mu(X)X^{-m})$ for every $m \geq 0$. By Leibniz's rule $\mathcal{S}(\mu_1)\mathcal{S}(\mu_2) \subset \mathcal{S}(\mu_1\mu_2)$, and $g \in \mathcal{S}(\mu)$ implies $g' \in \mathcal{S}(\mu/X)$ directly from the definition. We have $q_K \in \mathcal{S}(X^{1/3})$, and $1/\Psi \in \mathcal{S}(1/\log X)$: each derivative of $1/\Psi$ of order $m \geq 1$ is a sum of products $\Psi^{(j_1)} \dots \Psi^{(j_r)}/\Psi^{r+1}$ with $j_1 + \dots + j_r = m$, $j_i \geq 1$, hence $O(X^{-m}(\log X)^{-2})$. So $q_K^k/\Psi \in \mathcal{S}(X^{k/3}/\log X)$, and $g \mapsto \Psi^{-1}g'$ maps $\mathcal{S}(\mu)$ into $\mathcal{S}(\mu/(X \log X))$. The k -th term is therefore in $\mathcal{S}(X^{k/3}(X \log X)^{-(k-1)}(\log X)^{-1}) = \mathcal{S}(X^{1-2k/3}(\log X)^{-k})$.

For the leading part, write $q_K = 3\kappa X^{1/3} + r$ with $r \in \mathcal{S}(\log X)$. Then $q_K^k - (3\kappa)^k X^{k/3} \in \mathcal{S}(X^{(k-1)/3} \log X)$, and by the same calculus replacing q_K^k by $(3\kappa)^k X^{k/3}$ changes the k -th term by $\mathcal{S}(X^{2/3-2k/3}(\log X)^{1-k})$, a relative $O(X^{-1/3} \log X)$. In $(\Psi^{-1} \frac{d}{dX})^{k-1}(X^{k/3}/\Psi)$, expanded by Leibniz's rule, the summand in which every derivative falls on the power of X is $\Psi^{-k}(X^{k/3})^{(k-1)}$; every other summand contains a derivative of order $m \geq 1$ of some Ψ^{-j} , which lies in $\mathcal{S}(X^{-m}(\log X)^{-j-1})$, so it is in $\mathcal{S}(X^{1-2k/3}(\log X)^{-k-1})$. This gives the leading part stated in (iii), with relative error $O(1/\log X)$. For $k = 4$ the coefficient of $\kappa^4 X^{-5/3} \Psi^{-4}$ in it is $\frac{1}{24} \cdot 81 \cdot \frac{4}{3} \cdot \frac{1}{3} \cdot (-\frac{2}{3}) = -1$. For $k = 3$ it vanishes, and the first step gives the third term as $-\frac{9}{2} \kappa^3 \Psi^{-1}(\Psi^{-1}(X/\Psi)')' + O(X^{-4/3}(\log X)^{-2})$. Since $X\Psi' = 1 + O(1/X)$ and $X^2\Psi'' = -1 + O(1/X)$, direct differentiation gives $\Psi^{-1}(\Psi^{-1}(X/\Psi)')' = X^{-1}(-2\Psi^{-4} + 3\Psi^{-5}) + O(X^{-2}(\log X)^{-3})$, so the third term is $9\kappa^3 X^{-1} \Psi^{-4} - \frac{27}{2} \kappa^3 X^{-1} \Psi^{-5} + O(X^{-4/3}(\log X)^{-2})$. $\square$

(Dated note, 5 October 2026: item (iii) first ended “so successive terms decrease by factors $O(X^{-2/3}/\log X)$, and the third term is of the order of the remainder in (i)”, and Question 4 of Section 24.1 spoke of “the decrease of all terms”. Both statements are true of the term *bounds* and false of the terms themselves. By the last paragraph of the proof, the third term t_3 is $9\kappa^3 X^{-1} \Psi^{-4}(1 + O(1/\log X))$, smaller than its bound by a factor of order $1/\log X$, and the fourth is $t_4 = -\kappa^4 X^{-5/3} \Psi^{-4}(1 + o(1))$; so t_4/t_3 is of order $X^{-2/3}$, with no logarithmic gain. The independent check recorded at the end of Section 34 confirmed this numerically for $X = 10^6, \dots, 10^{20}$: $t_3 X \Psi^4$ runs from 10.8 to 15.49 (the two-term value above gives 15.48 at $X = 10^{20}$), $t_4 X^{5/3} \Psi^4 / (-\kappa^4)$ from 0.72 to 0.935, and $(t_4/t_3) X^{2/3} \log X$ grows in absolute value like $\log X$ (about -2.0 at $X = 10^6$ and -2.87 at $X = 10^{10}$). It also found that the remainder of (i) is t_3 up to a relative -2.7×10^{-8} at $X = 10^{10}$, so that numerically it is of order $X^{-1}(\log X)^{-4}$: the bound $O(X^{-1}(\log X)^{-3})$ in (i) is proved and correct, but one logarithm weaker than the truth. A proof of the sharper order would follow from the general truncation asked for in that Question. Item (i), item (ii) and the decay of the term bounds are unchanged.)

[Write note, 5 October 2026, batch 101.] Theorem 23.1 is the separation clause of the staircase theorem of the transseries volume (`p0:thm:staircase`) applied to the finite models f_K , as Part I's envelope (36) applies it to the smooth realization F ; Report 87 avoids constructing such a realization by working with f_K directly. The reversion (90) is Lagrange–Bürmann inversion in the gamma core $\log \Gamma(X+1) - X \log \rho$, where Part I inverts the cruder core $N \log(N/(e\rho))$ by Lambert's W (32); both are instances of `p0:thm:perturbed-inversion`. The source's claim that *every* finite truncation of (90) is justified is proved above only for the displayed truncation; the general case is recorded in Section 24.1. At intake the displayed formula was compared with x_3 at $y = \sqrt{a_{n-1}a_n}$, $n = 100, \dots, 600$: $[x_3] = n$ throughout, and the difference divided by $X^{-2/3}/\log X$ was 2.33–2.35.

24 Verification and further questions

The accompanying standard-library checker compares 71 terms of the new linear logarithm recurrence with the independent discrete insertion tree from [2] and with the published block recurrences from [3].

It also checks direct avoidance and initial-excedance enumeration through size eight, and 21 terms displayed in OEIS. A separate symbolic checker reconstructs $c_0, \dots, c_3$; it gives $c_2 = 0.981588612542\dots$ and $c_3 = -0.530220338435\dots$. Their exact expressions are included in `saddle_checks.json`. These finite checks support transcription and implementation; the enumeration and remainder theorems are proved above.

For example, after dividing the exact value of $a_n/n!$ by the leading factor in (71), the ratios at $n = 100, 200, 400, 600$ are respectively 0.95356734, 0.95755841, 0.96253451, 0.96559308. The approximations using $c_0, \dots, c_3$ are respectively 0.95445352, 0.95794679, 0.96270133, 0.96569406. This slow convergence helps explain why a fitted amplitude and power can differ from their limiting constants.

Three directions remain. First, a direct bijection explaining $A' = BA$ could expose a natural component decomposition and yield refined statistics. Second, the infinitely many singularities proved in Proposition 22.1 invite a systematic analysis of the corresponding exponentially smaller sectors, which is not supplied here. Third, marking the number of components in the exponential formula should give concentration and limit laws, but a combinatorial interpretation of those components should be settled before attributing such statistics to the original permutations.

This is an ordinary mathematical research note. Its proofs and executable checks were reviewed by direct derivation; it has not been externally refereed or formally verified.

[Write note, 5 October 2026, batch 101.] In this collection the checkers are shipped as `code/06-diagonal-check_exact.py`, `code/06-diagonal-check_saddle.py`, `code/06-diagonal-check_structure.py` and `code/06-diagonal-check_numerics.py`, with outputs `data/06-diagonal-exact_checks.json`, `data/06-diagonal-counts_600.json` (the counts through $n = 600$), `data/06-diagonal-saddle_checks.json`, `data/06-diagonal-structure_checks.json` and `data/06-diagonal-numeric_regression.json`; the “`saddle_checks.json`” of the text is the third of these. The scripts write beside themselves under the delivered names, so they are rerun only on a copy (see the README). At intake all four ran on a copy and reproduced the delivered outputs up to line endings. The ratios above agree with Part I’s table in Section 8 at $n = 100, 200, 400$.

24.1 Further questions and research

[Write note, 5 October 2026, batch 101.] Per the standing rule of this collection, the unproved items of Report 87 are collected here, each with its source and status.

1. *A direct bijection for $A' = BA$* (Report 87, first direction; Part I, Question 1). Part II’s separated cycles (Section 10.1) explain the exponential transform on the poset side; a direct map from the components to the initial-excedance permutations counted by B is not known. Open.
2. *The exponentially smaller sectors* (Report 87, second direction; Part I, Question 4). Answered for every fixed sector by Part IV (Theorems 28.1 and 30.1); the questions that remain (growing sector index, resurgence, certified values) are listed in Section 34.
3. *Limit laws for the number of components* (Report 87, third direction). The components of the exponential formula $A = e^T$ are, by Part II, separated cycles and singletons, so the interpretation the source asks for exists on the poset/word side; Part II marks the number of block pairs and of isolated elements, not the number of separated cycles, and identifies none of its statistics with a permutation statistic. Concentration and limit laws for the number of cycles are open.

4. *Every finite truncation of the reversion* (90) (Section 23; sketched there by “Taylor expansion of the finite models and the derivative estimate”). Proposition 23.2 proves the displayed two-term truncation and the decrease of all term bounds; a proof that the truncation after m terms has error of the order of the $(m + 1)$ -st term, for every m , is missing (it should follow from Lagrange inversion with remainder in a λ -deformation $G(x) = G(X) - \lambda q_K(x)$ and the symbol estimates of that proof, but this is not written).
5. *The farther chord parts* (Section 20). Stated without proof by the source; proved in Remark 20.1 (with an exponential loss in n). Not open.

No claim of Report 87 was found wrong. Its bibliography lists three works it does not cite ([9], [11], [12]); they are printed with the merged bibliography.

Part IV

Exact Hankel sectors at the singularities $\log 4 + 2\pi i k$ (Report 94)

[Write note, 5 October 2026, batch 101.] **Provenance.** This Part prints, in full, Report 94 of the session bundle of batch 101, *Canonical Exponentially Small Sectors for Vincular Avoidance*, dated 2 October 2026 (archive `Canonical_Vincular_Exponential_Sectors_Source.zip`, 384,547 bytes, arrival commit `60f54ea06`, placed in `f7c612c72`; its delivered PDF has 11 pages). Its author line reads “Research continuation of Exact Vincular Avoidance Asymptotics”; it carries no “prepared for private review” line. It records no repository pin: its only dependency is Report 87 (Part III here), identified by the SHA-256 of Report 87’s archive in `data/07-sectors-dependency.json`; at intake that digest was checked to be the digest of the delivered archive of Report 87. Report 94 saw neither Parts I–II nor the repository.

What it answers. It answers Part I’s Question 4 (“exponentially improved expansions accounting for continuation near the other singularities $\rho + 2\pi i k$ ”, Section 9) for every fixed sector: the coefficients $b_n = A_n/n!$ are the absolutely convergent sum of exact contour-defined sectors $H_k(n)$ (Theorem 28.1), and every fixed sector has a complete expansion (Theorem 30.1). Everything in this Part is new relative to Parts I–III except the $k = 0$ case of Theorem 30.1, which is Part I’s Theorem 1.1 for the exact dominant sector H_0 : $C_0 = D$, $\kappa_0 = c$, and the formulas (125)–(126) at $k = 0$ are (5)–(6) (checked symbolically at the write), as Report 94 itself says in Section 31. Its local branch at ζ_0 (Lemma 26.1) is (16)–(17), and its global branch (98) is Part I’s (2) written in $t = e^{z/2} = 2q$.

Numbering and editorial changes. Report 94’s Section n is Section $n + 24$, its equation (k) is equation ($k + 91$), and its Lemmas 2.1, 3.1, Theorem 4.1, Corollary 4.2 and Theorem 6.1 are Lemmas 26.1, 27.1, Theorem 28.1, Corollary 28.2 and Theorem 30.1. Its Appendix A is Appendix B (Part I keeps Appendix A), and its bibliography entry for Report 87 says so. Its 49 labels carry the prefix `vav:hk`; its citations of OEIS A113226, Bevan–Cheon–Kitaev and Elizalde point to Part I’s entries, and its entry for Report 87 is [13]. **One symbol is renamed:** the endpoint circle $C_k(r)$ of the source is printed $C_k(r)$, because the source also writes C_k for the amplitude (97) and C_r for a constant in Theorem 28.1. Its macros (`\Rea`, `\Ima`, `\atanh`, `\asinh`, `\acosh`, `\c0`, `\C`, `\Z`, `\R`) are kept; its `\dd` is Part I’s. No statement is changed. Write notes (dated) and one numbered write result (Remark 27.3, [write], after the source’s remark) are added.

Shipped records. The expanded proof note is `07-sectors-proof_notes.md`; the source comparison `07-sectors-source_status.md`; two internal audits

07-sectors-audit-global_hankel_review.md and
07-sectors-audit-sector_remainders_review.md; a rendering and replay record
07-sectors-qa-visual_review.md; the programs
code/07-sectors-check_sector_coefficients.py and
code/07-sectors-check_sector_numerics.py, with outputs
data/07-sectors-checks-symbolic_checks.json and
data/07-sectors-checks-numerical_checks.json. The audits are AI-internal; the Part is
unrefereed and not formalized.

[**Write note, 5 October 2026, batch 101.**] **Reading conventions of Part IV.** Report 94 keeps its own letters. Table 3 maps them to Parts I–III. The most tempting misreadings: $H_k(n)$ is a *number* (a sector of the coefficient), not Part I’s generating function $H(z)$; $b_n = a_n/n!$, not Part III’s A136127 counts; $g_k \in \{1, 2\}$ is a singular strength, not Part I’s Puiseux coefficient g_j ; and δ is a local variable at the singularity, not Part I’s $n^{-1/3}$.

Table 3: Reading conventions of Part IV (Report 94) against Parts I–III.

Part IV	meaning in Part IV	in Parts I–III (false reading)
a_n, b_n	counts; $b_n = a_n/n! = [z^n]A$	$A_n, A_n/n!$; Part III’s $b_n = \text{A136127}$
$A(z), A_k^\pm$	the EGF (global branch); bank values	Part I’s $H(z)$
$T, T_k, T_{0,k}$	$\log A$; local branch at ζ_k ; $\zeta_k/2 - 3$	Part I’s L ; $T_{0,0} = K$; Part I’s $T(x, y)$
$\mathcal{T}(u)$	local exponent of Section 30	not Part II’s $\mathcal{T} = \coth((\rho - w)/4)$
ζ_k, ϵ_k, g_k	$\rho + 2\pi i k$; $(-1)^k$; strength 1 or 2	Part II’s ζ is $z/\rho(u)$; Part I’s g_j are Puiseux coefficients
$\kappa_k, C_k, c_{k,j}$	sector constants	at $k = 0$: c, D, a_j (Part III: κ, C, c_j)
$H_k(n)$	the k -th sector (a number)	not Part I’s $H(z)$
$F(t), \Omega, D$	branch in $t = e^{z/2}$; t -plane domain; cut z -domain	$t = 2q$ with Part I’s q ; not Part I’s F, D , or Part II’s Ω
$q(x), L(x), B(x), B_*$	$\sqrt{1 - e^{-x}}$, $\operatorname{artanh} q$, bank and strip moduli	not Part I’s q, L ; not Part III’s $B(z)$
δ, δ_s	$\zeta_k - z$; the saddle point	Part I’s $\delta = n^{-1/3}$; Part III’s $\delta = \rho - z$ (the case $k = 0$)
r, r_n, R	radii	Part I’s $r_m, R(w)$
$C_k(r), C_r$	endpoint circle (renamed from $C_k(r)$); a constant	
M_r, M_*, Y_K, E_K	maxima; heights; rectangle error	Part I’s blocks M_i , threshold Y
$h, y, S(h, y)$	$n^{-1/6}$; Gaussian variable; formal exponent	h as in Parts I and III; Part II’s $S(w, u)$
$\phi, f_\phi, \alpha, c_k$	$\arg \zeta$; real phase; angle; a positive constant (Section 29)	Part I’s $\alpha = -1/3$; $c_k \neq c_{k,j}$
$a(\delta), b(\delta)$	analytic functions in the proof of Theorem 30.1	not the counts a_n, b_n

Abstract of Report 94. For the exponential generating function of permutations avoiding the vincular pattern 12–34, we construct an exact, absolutely convergent decomposition into full-cut Hankel sectors indexed by the singularities $\log 4 + 2\pi i k$. The clockwise endpoint loop is retained. This fixes an exact dominant sector and makes the first exponentially small conjugate pair unambiguous. Every fixed sector has a complete complex saddle expansion, with parity-dependent singular strength, explicit amplitude, and two correction coefficients. The proof uses the original horizontal cuts throughout; its complex Gaussian has positive real quadratic part. Exact symbolic checks, numerical regressions, scope of the source comparison, and reproducibility instructions are included.

25 Scope and main conclusions

Let a_n denote the number of permutations of $[n]$ avoiding the vincular pattern 12–34, and put

$$A(z) = \sum_{n \geq 0} \frac{a_n}{n!} z^n, \quad b_n = [z^n]A(z).$$

The identification is OEIS A113226 [1]. Our one mathematical dependency on the earlier delivered report 87, *Exact Vincular Avoidance Asymptotics* [13], is its exact generating function

$$A(z) = \exp T(z), \quad T(z) = \frac{z}{2} + \frac{\pi - 3 \arctan w}{w}, \quad w = \sqrt{4e^{-z} - 1}, \quad (92)$$

with the real positive square root for real $z < \rho := \log 4$. The earlier report proves its combinatorial origin and the complete dominant expansion. We do not repeat or claim those results as new here. Starting from (92), the analytic arguments below are self-contained.

Define

$$\zeta_k = \rho + 2\pi i k, \quad \epsilon_k = (-1)^k, \quad g_k = \frac{3 - \epsilon_k}{2} \in \{1, 2\}. \quad (93)$$

The main results are an exact normalization and its complete sector expansions:

$$b_n = \sum_{k \in \mathbb{Z}} H_k(n), \quad n \geq 1, \quad (94)$$

$$b_n = H_0(n) + 2 \operatorname{Re} H_1(n) + O_R(R^{-n}), \quad |\zeta_1| < R < |\zeta_2|, \quad (95)$$

$$H_k(n) \sim C_k \zeta_k^{-n} e^{3\kappa_k n^{1/3}} n^{-5/6}, \quad (96)$$

$$\kappa_k = \left(\frac{g_k^2 \pi^2}{4\zeta_k} \right)^{1/3}, \quad C_k = e^{\zeta_k/2-3} \sqrt{\frac{\kappa_k}{3\pi}}. \quad (97)$$

All roots in (97) are principal. The series in (94) is absolutely convergent, and H_k is defined by the complete contour in Section 27. Formula (96) extends to every algebraic order for each fixed k .

An all-orders formal expansion does not, by itself, determine an exponentially small correction. In (95), H_0 is an exact contour-defined quantity. Replacing it by a fixed finite dominant expansion would leave an algebraic error much larger than H_1 . Also, the real pair can exhibit oscillatory cancellation; our error estimates use absolute envelopes and do not assert a relative equivalent at its near-zeros.

[Write note, 5 October 2026, batch 101.] “The earlier delivered report 87” is Part III of this report, and the generating function (92) is also Part I’s Theorem 1.1, proved independently there; so the one input of this Part is proved twice in this report. In Part I’s letters, (96) at $k = 0$ reads $H_0(n) \sim D\rho^{-n} e^{3cn^{1/3}} n^{-5/6}$, the right side of (4) divided by $n!$; for $k \neq 0$ the ratio $|H_k(n)/H_0(n)|$ decays like $(\rho/|\zeta_k|)^n$ up to stretched factors, which is why these sectors are invisible in Part I’s Poincaré expansion.

26 A global branch and the parity of local strengths

Let

$$\Omega = \mathbb{C} \setminus ((-\infty, -2] \cup [2, \infty)).$$

On Ω , use the branches of $\sqrt{4-t^2}$ and $\arcsin(t/2)$ with values 2 and 0 at $t = 0$. Set

$$F(t) = t \exp \left\{ \frac{(3 \arcsin(t/2) - \pi/2)t}{\sqrt{4-t^2}} \right\}, \quad A(z) = F(e^{z/2}). \quad (98)$$

The preimage of the two t -cuts is exactly the family

$$\zeta_k + [0, \infty), \quad k \in \mathbb{Z}.$$

Thus (98) gives a single-valued holomorphic branch on their complement D . On the real interval $z < \rho$, the identity

$$\arcsin(e^{z/2}/2) = \frac{\pi}{2} - \arctan \sqrt{4e^{-z} - 1}$$

proves agreement with (92); the identity theorem then identifies the continuation.

Oddness of $\arcsin$ and evenness of the normalized square root give

$$F(-t) = -F(t) \exp \left(\frac{\pi t}{\sqrt{4-t^2}} \right). \quad (99)$$

In particular $A(z + 4\pi i) = A(z)$, whereas a $2\pi i$ shift is not a period. The branch is real under conjugation: $A(\bar{z}) = \overline{A(z)}$.

Lemma 26.1 (Exact local branch). *For sufficiently small $\delta \in \mathbb{C} \setminus (-\infty, 0]$, with principal $\sqrt{\delta}$, one has $A(\zeta_k - \delta) = \exp T_k(\zeta_k - \delta)$, where*

$$T_k(\zeta_k - \delta) = \frac{\zeta_k - \delta}{2} + \frac{g_k \pi}{\sqrt{e^\delta - 1}} - 3 \frac{\arctan \sqrt{e^\delta - 1}}{\sqrt{e^\delta - 1}}. \quad (100)$$

Consequently, with $T_{0,k} = \zeta_k/2 - 3$,

$$T_k(\zeta_k - \delta) = g_k \pi \delta^{-1/2} + T_{0,k} - \frac{g_k \pi}{4} \delta^{1/2} + \frac{\delta}{2} + O(\delta^{3/2}). \quad (101)$$

Proof. Put $w = \sqrt{e^\delta - 1}$. For even k , $t = e^{(\zeta_k - \delta)/2} = 2e^{-\delta/2}$ and $\arcsin(t/2) = \pi/2 - \arctan w$. For odd k , $t = -2e^{-\delta/2}$ and $\arcsin(t/2) = -\pi/2 + \arctan w$, while $t/\sqrt{4-t^2} = -1/w$. Substitution gives (100). The quotient $\arctan w/w$ is analytic in w^2 near zero. Expanding it and $(e^\delta - 1)^{-1/2}$ proves (101). The logarithm is fixed modulo $2\pi i$; its exponential retains the important factor $e^{\zeta_k/2} = 2(-1)^k$. $\square$

[Write note, 5 October 2026, batch 101.] Part I's Section 4 continues the EGF only in a disc of radius less than $|\zeta_1|$, cut along $[\rho, R)$, and notes that the ODE continues it along paths avoiding $\rho + 2\pi i k$. The global branch (98) on the domain D with horizontal cuts is new here; it is “canonical” only in the sense the source states in Section 27 (this branch, these cuts). By Lemma 26.1 every ζ_k is a genuine essential singularity of this branch, in agreement with Proposition 22.1 of Part III (see the note after it); the $k = 0$ case of (100) is Part I's (16) with $w = \delta$ there.

27 Bank values and the full-cut sectors

For $x > 0$, write

$$q(x) = \sqrt{1 - e^{-x}}, \quad L(x) = \operatorname{artanh} q(x) = \operatorname{arcosh}(e^{x/2}), \quad B(x) = 2e^{x/2} e^{-3L(x)/q(x)}. \quad (102)$$

Denote the upper and lower limits on the k th cut by $A_k^+(x)$ and $A_k^-(x)$.

Lemma 27.1 (Banks and their bounds). *For every $k \in \mathbb{Z}$ and $x > 0$,*

$$A_k^\pm(x) = \epsilon_k B(x) \exp\left(\pm \frac{ig_k \pi}{q(x)}\right). \quad (103)$$

Moreover,

$$B(x) \longrightarrow 2e^{-3} \quad (x \downarrow 0), \quad B(x) = \frac{1}{4}e^{-x}\{1 + O(xe^{-x})\} \quad (x \rightarrow \infty). \quad (104)$$

Thus B is bounded and integrable on $(0, \infty)$.

Proof. On the upper bank, $\delta = -x - i0$, so the local square root continues as $-iq$. Also $\arctan(-iq) = -i \operatorname{artanh} q$. The lower bank has the conjugate choices. This gives (103), first near the endpoint and then along the complete bank by continuation. As $x \downarrow 0$, $L/q \rightarrow 1$. At infinity, $q = 1 + O(e^{-x})$ and $L = x/2 + \log 2 + O(e^{-x})$, giving (104). In fact both individual banks have leading value $-e^{-x}/4$ at infinity. $\square$

Fix

$$0 < r < \min(\rho/2, \pi). \quad (105)$$

Let $\mathcal{C}_k(r)$ be the endpoint circle traversed clockwise from the lower bank of $\zeta_k + r$ to its upper bank. Equivalently,

$$z = \zeta_k - re^{i\alpha}, \quad \alpha : \pi \longrightarrow -\pi.$$

For integer $n \geq 1$, define the *canonical full-cut sector*

$$H_k(n) = \frac{1}{2\pi i} \left\{ \int_{\mathcal{C}_k(r)} \frac{A(z)}{z^{n+1}} dz + \int_r^\infty \frac{A_k^+(x) - A_k^-(x)}{(\zeta_k + x)^{n+1}} dx \right\}. \quad (106)$$

The path runs inward along the lower bank, clockwise around the endpoint, and outward along the upper bank. A pole test fixes the sign: with $A(z) = 1/(\rho - z)$, a clockwise circle contributes $-\operatorname{Res}_{z=\rho}(A(z)z^{-n-1}) = \rho^{-n-1}$.

Cauchy's theorem in the cut annulus between two radii satisfying (105) proves that (106) is independent of r . Accordingly, the radius may later depend on n . The word “canonical” here refers to the specified global branch and its horizontal cuts.

Remark 27.2 (Why the endpoint loop matters). The jump integral in (106) converges absolutely even if its lower limit is replaced by zero, because B is bounded. The problem with a jump-only formula is therefore not divergence. It is the omission of the endpoint loop, whose integrand grows like $\exp(g_k \pi / \sqrt{\delta})$ on the left of the endpoint. There is no vanishing small-circle estimate. Radius independence and integrable bank values imply that the shrinking loop has a finite integral limit, but do not make that limit zero. Retaining a fixed positive radius is sufficient for all our arguments.

Remark 27.3 ([write] An explicit majorant for the banks). For every $x > 0$, $B(x) < 2e^{-x}(1+q(x))^{-3} \leq 2e^{-x}$; in particular $\int_0^\infty B(x) dx < 2$. Indeed $q \in (0, 1)$ and $(1+q)/(1-q) = (1+q)^2/(1-q^2) = (1+q)^2 e^x$, so $L = \operatorname{artanh} q = \frac{1}{2} \log \frac{1+q}{1-q} = \frac{x}{2} + \log(1+q) > 0$. Hence $L/q > L$ and $3L/q > \frac{3x}{2} + 3\log(1+q)$, which gives the bound. The first bound is asymptotically sharp, both sides being $\sim e^{-x}/4$ as $x \rightarrow \infty$ by (104). The same identity gives $L = \operatorname{arcosh}(e^{x/2})$ and the limit $L = x/2 + \log 2 + O(e^{-x})$ used in Lemma 27.1. The bound $B \leq 2e^{-x}$ is the one the delivered `code/07-sectors-check_sector_numerics.py` uses, in a comment without proof, for its omitted-bank tail bounds.

28 Exact decomposition and uniform sector tails

Theorem 28.1 (Absolutely convergent sector decomposition). *For every integer $n \geq 1$,*

$$b_n = \sum_{k \in \mathbb{Z}} H_k(n), \quad H_{-k}(n) = \overline{H_k(n)}. \quad (107)$$

For every fixed admissible r , there is $C_r < \infty$, independent of k, n , such that

$$|H_k(n)| \leq C_r (|\zeta_k| - r)^{-n-1}. \quad (108)$$

The series in (107) is therefore absolutely convergent.

Proof. Periodicity leaves only two translated endpoint-circle branches. Let M_r be the maximum of $|A|$ on these two compact circle banks. Since $\operatorname{Re} \zeta_k = \rho > 0$,

$$|\zeta_k + x| \geq |\zeta_k| \quad (x \geq 0), \quad |z| \geq |\zeta_k| - r \quad (z \in \mathcal{C}_k(r)).$$

Thus (108) follows with

$$C_r = rM_r + \frac{1}{\pi} \int_r^\infty B(x) dx.$$

Since $|\zeta_k| \sim 2\pi|k|$, the bound implies absolute convergence for $n \geq 1$.

It remains to prove the exact identity. On the lines $\operatorname{Im} z = (2j+1)\pi$, write $s = e^{x/2}$. Formula (98) gives the common modulus

$$B_*(x) = s \exp \left(-\frac{3s \operatorname{arsinh}(s/2)}{\sqrt{4+s^2}} \right). \quad (109)$$

It is $O(e^{x/2})$ as $x \rightarrow -\infty$ and $O(e^{-x})$ as $x \rightarrow \infty$, so $M_* := \int_{\mathbb{R}} B_*(x) dx < \infty$.

The vertical sides also vanish. At zero, $F(t) = t(1 + O(t))$. At infinity, uniformly in the closed upper or lower half-plane with the respective bank limits,

$$F(t) = -t^{-2} \left\{ 1 + O \left(\frac{\log |t|}{|t|^2} \right) \right\}. \quad (110)$$

For example, in the upper half-plane, $\arcsin(t/2) = \pi/2 + i \log t + O(t^{-2})$ and $t/\sqrt{4-t^2} = i + O(t^{-2})$; the lower half-plane is analogous. These expansions also hold in the boundary limits. Hence $A(z) = O(e^{\operatorname{Re} z/2})$ on left vertical sides and $O(e^{-\operatorname{Re} z})$ on right vertical sides.

Fix $K \geq 0$, set $Y_K = (2K+1)\pi$, and apply the residue theorem to $A(z)z^{-n-1}$ in the rectangle with horizontal sides at $\pm Y_K$, slit along the cuts $|k| \leq K$ and with radius- r endpoint disks removed. The external rectangle is counterclockwise; the slit boundaries are clockwise with the analytic domain to the left. Each is lower bank inward, clockwise endpoint loop, upper bank outward, exactly the orientation of (106). Send the two vertical sides to infinity. Their integrals vanish, and the bank integrals converge absolutely. The only enclosed pole is at zero, with residue b_n . Thus

$$b_n = \sum_{|k| \leq K} H_k(n) + E_K(n), \quad |E_K(n)| \leq \frac{M_*}{\pi Y_K^{n+1}}. \quad (111)$$

Letting $K \rightarrow \infty$ proves (107). Reflection of the entire oriented contour, including $1/(2\pi i)$, proves conjugacy. $\square$

Corollary 28.2 (Exact dominant normalization). For $|\zeta_1| < R < |\zeta_2|$,

$$b_n = H_0(n) + 2 \operatorname{Re} H_1(n) + O_R(R^{-n}). \quad (112)$$

More generally, a fixed number of sectors may be retained, with a geometric remainder separated from the next singularity modulus.

Proof. Choose r still smaller so that $R < |\zeta_2| - r$. For $n \geq 1$, factor R^{-n} in (108) and use

$$\left(\frac{R}{|\zeta_k| - r} \right)^n \leq \frac{R}{|\zeta_k| - r}, \quad |k| \geq 2.$$

The remaining inverse-square sum converges. Conjugacy gives (112). The same argument applies after any fixed finite collection of nearest cuts. $\square$

[Write note, 5 October 2026, batch 101.] This theorem and its corollary are the exponentially improved form asked for in Part I's Question 4, and are new. The delivered numerical checks test each H_k against its own asymptotics (Section 32), not the sum (107). The sum itself was checked independently at intake, by contour quadrature of $H_k(n)$ for $0 \leq k \leq 6$ (radius 0.5, 40 digits): with b_n from the exact recurrence, the truncated sum $H_0 + 2 \sum_{k=1}^6 \operatorname{Re} H_k$ agreed with b_n to a relative error 3.5×10^{-12} at $n = 6$ (where the two last sectors still have size $|H_5| + |H_6| = 3.4 \times 10^{-11}$), 3.5×10^{-26} at $n = 15$ and 1.5×10^{-39} at $n = 30$; $H_0(n)$ was real, and at $n = 30$, $H_0/b_n - 1 = -7.5175154 \times 10^{-22} = -2 \operatorname{Re} H_1/b_n$. These are numerical regressions, not certificates.

29 The complex saddle on the horizontal endpoint circle

Fix k for this section, and abbreviate $\zeta = \zeta_k$, $g = g_k$, $\phi = \arg \zeta$, $\kappa = \kappa_k$. Then

$$|\phi| < \frac{\pi}{2}, \quad \arg \kappa = -\frac{\phi}{3}, \quad \operatorname{Re} \kappa > 0.$$

Use radius

$$r_n = \left| \left(\frac{g\pi\zeta}{2n} \right)^{2/3} \right|, \quad \delta_s = \zeta \kappa n^{-2/3} = r_n e^{2i\phi/3}. \quad (113)$$

For large n this is admissible in (106). On $\delta = r_n e^{i\alpha}$, $-\pi \leq \alpha \leq \pi$, the exponent, after removing the constant and ζ^{-n-1} , is

$$\frac{n\delta}{\zeta} + \frac{g\pi}{\sqrt{\delta}} + O(n^{-1/3}), \quad (114)$$

uniformly on the whole circle. Indeed, the local analytic correction is $O(\sqrt{r_n})$ and the quadratic kernel correction is $O(nr_n^2)$.

The leading real part of (114) is

$$|\kappa| n^{1/3} f_\phi(\alpha), \quad f_\phi(\alpha) = \cos(\alpha - \phi) + 2 \cos(\alpha/2). \quad (115)$$

Its derivative factors as

$$f'_\phi(\alpha) = -2 \sin(3\alpha/4 - \phi/2) \cos(\alpha/4 - \phi/2). \quad (116)$$

Throughout the angular interval the cosine is positive, and the sine argument lies strictly between $-\pi$ and π . Hence the unique global maximum is

$$\alpha_s = 2\phi/3, \quad f_\phi(\alpha_s) = 3 \cos(\phi/3).$$

This proves the principal saddle selection without rotating any cut.

Writing $\alpha = \alpha_s + \theta$, the leading complex phase becomes

$$\kappa n^{1/3} \left(e^{i\theta} + 2e^{-i\theta/2} \right), \quad (117)$$

with second derivative $-3\kappa n^{1/3}/2$ at zero. Since $\text{Re } \kappa > 0$, the real θ contour already has a decaying complex Gaussian. For some $c_k > 0$, the real-part deficit is bounded below by

$$c_k n^{1/3} \min(\theta^2, 1)$$

on the fixed angular interval. In particular, angles outside $|\theta| \leq n^{-1/6+\eta}$ have relative loss $e^{-c_k n^{2\eta}}$, for any fixed sufficiently small $\eta > 0$.

The bank integrals are exponentially smaller. Lemma 27.1 and monotonicity of $|\zeta + x|$ for $x \geq 0$ give

$$\begin{aligned} \int_{r_n}^{\infty} B(x) |\zeta + x|^{-n-1} dx &\leq \|B\|_1 |\zeta + r_n|^{-n-1} \\ &= O_k \left(|\zeta|^{-n-1} e^{-|\kappa| \cos \phi n^{1/3} + O(n^{-1/3})} \right). \end{aligned} \quad (118)$$

This has strictly smaller exponential rate than the saddle envelope $|\zeta|^{-n} e^{3 \text{Re } \kappa n^{1/3}}$.

Finally the clockwise loop, with the orientation reversed after parametrization, is exactly

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} r_n e^{i\alpha} A(\zeta - r_n e^{i\alpha}) (\zeta - r_n e^{i\alpha})^{-n-1} d\alpha. \quad (119)$$

At α_s the numerator is δ_s , and $\delta_s/\zeta = \kappa n^{-2/3}$. Gaussian evaluation yields

$$e^{\zeta/2-3} \zeta^{-n} e^{3\kappa n^{1/3}} \frac{\kappa n^{-2/3}}{2\pi} \sqrt{\frac{4\pi}{3\kappa n^{1/3}}} = C_k \zeta_k^{-n} e^{3\kappa_k n^{1/3}} n^{-5/6}.$$

The negative sign of $e^{\zeta_k/2}$ for odd k is therefore part of the amplitude, rather than an arbitrary phase convention.

[Write note, 5 October 2026, batch 101.] The positive constant c_k of this section is not a coefficient $c_{k,j}$ of Section 30. At $k = 0$ the saddle circle is centred at ρ with radius $r_n = \rho c n^{-2/3}$ and $\delta_s = r_n$, so the saddle point $\rho - \delta_s = \rho(1 - cn^{-2/3})$ is Part I's $\rho(1 - t_0)$ and Part III's chord abscissa; Part I and Part III deform to a vertical segment there, while this Part stays on the circle. The internal audit `07-sectors-audit-sector_remainders_review.md` notes that its window exponent η here equals $\eta/6$ in Section 30's h -based window, a reuse of notation.

30 All orders and explicit correction coefficients

The circle formulation also provides a finite coefficient algorithm directly on the original contour. Put

$$h = n^{-1/6}, \quad \alpha = \alpha_s + hy, \quad u(h, y) = \kappa h^4 e^{ihy}, \quad T_0 = \zeta/2 - 3,$$

and define

$$\mathcal{T}(u) = \frac{\zeta(1-u)}{2} + \frac{g\pi}{\sqrt{e^{\zeta u} - 1}} - 3 \frac{\arctan \sqrt{e^{\zeta u} - 1}}{\sqrt{e^{\zeta u} - 1}}, \quad (120)$$

$$\begin{aligned} S(h, y) &= \mathcal{T}(u(h, y)) - (h^{-6} + 1) \log(1 - u(h, y)) \\ &\quad - 3\kappa h^{-2} - T_0 + \frac{3\kappa}{4} y^2 + ihy. \end{aligned} \quad (121)$$

The last term is the logarithm of the circle Jacobian factor e^{ihy} . The relation $g\pi/\sqrt{\zeta\kappa} = 2\kappa$ on the chosen branches cancels all negative powers in (121); the constant term also cancels. Thus

$$S(h, y) \in h\mathbb{C}[y][[h]],$$

and the exact formal identity $S(-h, -y) = S(h, y)$ shows that its coefficient of h^j has parity j as a polynomial in y .

Theorem 30.1 (Complete expansion of each fixed sector). *For every fixed $k \in \mathbb{Z}$ and $K \geq 0$,*

$$H_k(n) = C_k \zeta_k^{-n} e^{3\kappa_k n^{1/3}} n^{-5/6} \left\{ \sum_{j=0}^K c_{k,j} n^{-j/3} + O_{k,K}(n^{-(K+1)/3}) \right\}. \quad (122)$$

The coefficients are determined by

$$c_{k,j} = \sqrt{\frac{3\kappa}{4\pi}} \int_{\mathbb{R}} e^{-3\kappa y^2/4} [h^{2j}] e^{S(h,y)} dy. \quad (123)$$

In particular,

$$c_{k,0} = 1, \quad (124)$$

$$c_{k,1} = \frac{\kappa^2(1-\zeta)}{2} - \frac{5}{36\kappa}, \quad (125)$$

$$c_{k,2} = \frac{\kappa^4(1-\zeta)^2}{8} + \frac{\kappa(53\zeta-17)}{72} - \frac{35}{2592\kappa^2}. \quad (126)$$

Proof. On the central window $|y| \leq h^{-\eta}$, with a fixed $0 < \eta < 1$, the circle contribution equals

$$e^{T_0} \zeta^{-n} e^{3\kappa h^{-2}} \frac{\kappa h^5}{2\pi} \int_{-h^{-\eta}}^{h^{-\eta}} e^{-3\kappa y^2/4} e^{S(h,y)} dy. \quad (127)$$

For every finite order J , Taylor expansion of the convergent expression (120), the logarithm, and the exponential has a remainder bounded by

$$C_J h^{J+1} (1 + |y|^{m_J}) \exp\{C_J h(1 + |y|^3)\}$$

for some finite m_J . For detail, write $a(\delta) = \sqrt{\delta/(e^\delta - 1)}$ with $a(0) = 1$, and $b(\delta) = -\delta/2 - 3(\arctan \sqrt{e^\delta - 1}/\sqrt{e^\delta - 1} - 1)$. Both are analytic at zero. The nonconstant part of $\mathcal{T}$ is $2\kappa h^{-2} e^{-ihy/2} a(\zeta \kappa h^4 e^{ihy}) + b(\zeta \kappa h^4 e^{ihy})$. After cancellation of the negative powers, the leading residual phase is cubic in y times h . On $0 \leq t \leq h$, every fixed t -derivative of $S(t, y)$ is polynomially bounded in $|y|$, and $|S(t, y)| \leq Ct(1 + |y|^3)$ because $|ty| \leq h^{1-\eta}$. Taylor's theorem and the finite derivative formulas for e^S give the displayed bound. For $h|y| \rightarrow 0$, its last exponential is absorbed by a smaller fixed Gaussian because $\text{Re } \kappa > 0$. The integrated remainder is consequently $O(h^{J+1})$.

Expand through $J = 2K + 1$. Odd polynomial coefficients integrate to zero on the symmetric window. Extending each polynomial–Gaussian integral to $\mathbb{R}$ costs less than every power of h . The angular minor-arc estimate in Section 29 controls the omitted circle, and (118) controls the two banks. This proves an integrated error $O(h^{2K+2})$ relative to the leading Gaussian factor. Since

$$\int_{\mathbb{R}} e^{-3\kappa y^2/4} dy = \sqrt{\frac{4\pi}{3\kappa}},$$

with principal root, the prefactor in (127) is exactly $C_k n^{-5/6}$. This proves (122) and (123).

The normalized moments are

$$\mathbb{E}(y^{2m}) = (2m-1)!! \left(\frac{2}{3\kappa}\right)^m, \quad \mathbb{E}(y^{2m+1}) = 0. \quad (128)$$

Only finitely many Taylor terms are used at each order, so (123) is an algebraic algorithm. The first two terms of the exponent are

$$S = i \left(y - \frac{\kappa y^3}{8} \right) h + \left(\frac{\kappa^2(1-\zeta)}{2} + \frac{3\kappa y^4}{64} \right) h^2 + O(h^3). \quad (129)$$

Taking the moments of $[h^2]e^S$ gives (125). For (126), all required terms are generated by

$$\begin{aligned} S &= \frac{\kappa}{h^2} (e^{ihy} + 2e^{-ihy/2} - 3) + \frac{3\kappa y^2}{4} + ihy \\ &\quad + \frac{\kappa^2 h^2}{2} (e^{2ihy} - \zeta e^{ihy/2}) + \kappa(1 + \zeta/2) h^4 e^{ihy} + O(h^6), \end{aligned} \quad (130)$$

where the displayed exponentials are expanded through degree four. Applying (128) to $[h^4]e^S$ yields (126). The supplied symbolic script verifies both formulas exactly. $\square$

Remark 30.2 (Dependence on the sector index). All constants in the local asymptotic proof may depend on fixed k . No interchange of the $k \rightarrow \infty$ and $n \rightarrow \infty$ limits is claimed. Uniform control of the infinite remainder instead comes from the exact fixed-radius bound (108). This separates the two roles cleanly.

[**Write note, 5 October 2026, batch 101.**] At $k = 0$ ($\zeta = \rho$, $\kappa = c$) the formulas (125) and (126) are Part I's (5) and (6); the identity of (126) with (6) was checked symbolically at the write. The internal audit `07-sectors-audit-sector_remainders_review.md` also lists the next two exponent coefficients, $S_3 = i\kappa^2 y(1 - \zeta/4) + i\kappa y^5/128$ and $S_4 = \kappa(1 + \zeta/2) + \kappa^2 y^2(\zeta/16 - 1) - 11\kappa y^6/7680$, and recovers $c_{k,1}$, $c_{k,2}$ from them independently. Values not printed by the source, computed at the write from (125)–(126):
 $c_{1,2} = 6.194849135854 \dots + 12.60843584359 \dots i$,
 $c_{2,1} = -1.996687241967 \dots - 1.24647320807 \dots i$, $\kappa_2 = 0.512623514249 \dots - 0.271438168874 \dots i$
and $C_2 = 0.023974071910 \dots - 0.005955539234 \dots i$.

31 The first conjugate pair with an absolute error

Let

$$\mathcal{E}_1(n) = |\zeta_1|^{-n} e^{3 \operatorname{Re} \kappa_1 n^{1/3}} n^{-5/6}.$$

Combining Corollary 28.2 and Theorem 30.1 gives, for every fixed $K \geq 0$ and $|\zeta_1| < R < |\zeta_2|$,

$$b_n - H_0(n) = 2 \operatorname{Re} \left\{ C_1 \zeta_1^{-n} e^{3\kappa_1 n^{1/3}} n^{-5/6} \sum_{j=0}^K c_{1,j} n^{-j/3} \right\} + O_K(\mathcal{E}_1(n) n^{-(K+1)/3}) + O_R(R^{-n}). \quad (131)$$

The leading oscillatory factor can equivalently be written

$$2|C_1| \mathcal{E}_1(n) \cos(-n \arg \zeta_1 + 3 \operatorname{Im} \kappa_1 n^{1/3} + \arg C_1).$$

Formula (131), rather than an equivalent for this cosine, remains valid near cancellation indices.

For numerical orientation,

$$\begin{aligned} |\zeta_1| &= 6.434301023424 \dots, \\ |\zeta_2| &= 12.642605841879 \dots, \\ \kappa_1 &= 1.037853533783 \dots - 0.502894353735 \dots i, \\ C_1 &= -0.033949227189 \dots + 0.007791818790 \dots i, \\ c_{1,1} &= -3.546963506118 \dots - 2.440313169662 \dots i. \end{aligned}$$

At $k = 0$, the leading constant is $0.035706041258 \dots$ and $c_{0,1} = -0.398274243295 \dots$, agreeing with report 87. The exact difference $b_n - H_0(n)$ is exponentially smaller than the dominant scale, so uniqueness of Poincare coefficients also identifies the complete $k = 0$ expansion with that earlier one.

[Write note, 5 October 2026, batch 101.] These constants were recomputed at the write and at intake; all printed digits agree. The $k = 0$ values are Part I's D and a_1 (Section 1 and the table of Section 6), and the uniqueness argument of the last sentence identifies $c_{0,j}$ with Part I's a_j (and Part III's c_j) for every j . Relative to the dominant sector, the first pair is smaller by about $(\rho/|\zeta_1|)^n e^{-3(c - \operatorname{Re} \kappa_1)n^{1/3}}$, with $\rho/|\zeta_1| = 0.2155 \dots$ and $c - \operatorname{Re} \kappa_1 = 0.1740 \dots$

32 Exact checks and clean replay

The source bundle includes the following independently runnable checks.

1. `check_sector_coefficients.py` computes the circle exponent through fourth order, performs exact Gaussian polynomial moments, and verifies c_0, c_1, c_2 using symbolic arithmetic. An independent straight-chord Gaussian coordinate reproduces the same coefficients. It also checks the parity cancellations and the odd-shift exponent identity. Acceptance tests use explicit exceptions, so Python optimization does not disable them.
2. `check_sector_numerics.py` evaluates the endpoint circle and both banks at high precision for $k = 0, 1, 2$ and $n = 512, 4096$. It reports the normalized sector and the n -scaled remainder after c_2 . These are numerical regressions, not interval certificates or substitutes for the proofs.
3. `replay.sh` runs both checks in a fresh output directory, under `python -O`, then compiles the article twice. `build.sh` provides the shorter PDF-only build. The exact input ZIP is identified by SHA-256 in `dependency.json`; it is not overwritten or repackaged as a new result.

For example, if $Q_k(n)$ is the ratio of $H_k(n)$ to its leading complex term, direct numerical integration gives the values below. The last column is $n\{Q_k(n) - 1 - c_{k,1}n^{-1/3} - c_{k,2}n^{-2/3}\}$.

k	n	$Q_k(n)$	Scaled remainder
0	512	0.964394490078	−0.593178410
0	4096	0.978803891189	−0.566473208
1	512	$0.651554305673 - 0.165541159156i$	$-0.957324191 - 29.44451738i$
1	4096	$0.802730333675 - 0.111269474246i$	$0.888518127 - 32.77456858i$
2	512	$0.806825329379 - 0.083265991203i$	$-1.838945967 - 15.02458344i$
2	4096	$0.889923426656 - 0.056494645833i$	$-1.164706665 - 16.63829059i$

The finite data are consistent with the proven $O(n^{-1})$ normalized error after c_2 . The limiting third coefficient is not estimated or asserted from this table.

[Write note, 5 October 2026, batch 101.] In this collection the two checks are shipped as `code/07-sectors-check_sector_coefficients.py` and `code/07-sectors-check_sector_numerics.py`, the replay and build scripts as `code/07-sectors-replay.sh` and `code/07-sectors-build.sh`, and the outputs and dependency record as `data/07-sectors-checks-symbolic_checks.json`, `data/07-sectors-checks-numerical_checks.json` and `data/07-sectors-dependency.json`. The two checks write `checks/*.json` relative to the working directory; the replay and build scripts need the delivered `article.tex`, which is not shipped (this article replaces it). At intake both checks ran on a copy and reproduced the delivered JSON outputs up to line endings. The table above agrees with `data/07-sectors-checks-numerical_checks.json` to the digits printed.

33 Source comparison and the bounded novelty claim

The source comparison was refreshed on 2 October 2026. The inspected OEIS entry [1], revision 49 dated 5 November 2025, supplies the sequence definition and recursive enumeration. Bevan, Cheon, and Kitaev [2] provide the insertion-tree framework and state the dominant growth and stretched-exponential form as Conjectures 13 and 14. Elizalde [3], Proposition 5.1, supplies earlier coefficientwise generating-function bounds. These are foundational sources; this continuation does not relabel their work.

The frozen report 87 [13] already establishes the exact EGF, dominant all-orders expansion, inversion bounds, differential algebraicity, and non-D-finiteness. The new result relative to that supplied baseline is the exact full-cut sector decomposition, the parity-sensitive complex sectors with remainder control, and the resulting canonically normalized exponentially small pair. None of those assertions is needed to re-prove the earlier combinatorial identity.

Targeted searches combining A113226 or 12–34 avoidance with “Hankel”, “arctan”, “transseries”, and exponentially small asymptotics did not locate the sector result proved here. This is a bounded search statement, not a worldwide priority determination. In particular, a failure of a search engine to return a match is not evidence of absence from all literature. The contour and saddle methods themselves are classical. The report is a mathematical research argument with reproducible checks, not a claim of external peer review or formal proof verification.

[Write note, 5 October 2026, batch 101.] Report 94 inspected OEIS A113226 on 2 October 2026 at revision 49 (<https://oeis.org/A113226/internal>); the merged entry [1] records Part I’s inspection of 1 October. Its copy of [2] cites the author-institution record <https://strathprints.strath.ac.uk/91802/> and its copy of [3] the DOI 10.1016/j.aam.2005.05.006. Relative to this report, the baseline is Parts I and III, which both establish the dominant results; the incremental claim stands unchanged against Parts I–III, none of which treats the sectors $k \neq 0$.

34 Further questions with specific next steps

[Write note, 5 October 2026, batch 101.] This section serves as Part IV’s “Further questions and research” section in the sense of the collection’s standing rule; Report 94 states no unproved claim outside it, and none of its claims was found wrong. The first question below refines Part I’s Question 4, which Part IV answers for each fixed k ; the third is a case of Part I’s Question 3 (certified constants), still open. Report 87’s open items are in Section 24.1. A further question, not asked by any source: Part II’s marked generating function $H(z, u)$ has, for $u \neq 1$, its own complex singularities, and its sectors are not treated (Part II excludes them, Section 16).

Uniformly growing sector index. The present local expansion fixes k , while the exact tail estimate is uniform. A useful next problem is a simultaneous expansion when $|k|$ grows with n . Since $|\delta_s| \asymp (|k|/n)^{2/3}$, the small-neighborhood requirement naturally suggests first treating $|k| = o(n)$ and tracking the angular-gap constants. This scaling observation is motivation, not a uniform theorem.

Late coefficients and optimal truncation. Determine large- j growth of $c_{0,j}$ and whether an optimally truncated dominant expansion can be connected quantitatively to the exact H_0 . Such a result would relate the contour normalization to a resummation procedure. No Borel summability or resurgence claim follows merely from the finite-order results proved here.

Certified numerical extraction. Interval quadrature of (106), using explicit cut-bank tails and a saddle-centered circle, could produce certified values for exponentially small sectors without subtracting two enormous nearly equal dominant quantities. A useful implementation target is a verified absolute error below the $|k| \geq 2$ tail in (112).

Oscillatory cancellation. The phase in (131) describes the first pair’s sign changes and near-zeros. Establishing quantitative lower bounds away from zero on selected integer subsequences would require control of the arithmetic sampling of this phase. The current absolute-error formulation deliberately avoids such an unproved lower bound.

Other combinatorial transforms. The key global feature is a periodic exponential pullback with two different singular strengths. It is natural to seek other exact combinatorial EGFs with this structure and to test whether the same full-cut normalization supplies their beyond-all-orders terms. An application must verify its global branch and boundary decay separately; local singular expansions alone are insufficient.

Independent check of the write (dated note, 5 October 2026). An independent adversarial check of the batch-101 write of Parts III and IV (07fd071f8), made by the intake afterwards, reread the four write results, Remarks 19.1, 20.1 and 27.3 and Proposition 23.2, and spot-checked Part IV’s text, constants and table against the delivered manuscript. It found the three remarks and Part IV valid, and Proposition 23.2 valid except for one sentence of item (iii), which is false as written: the *bounds* for successive reversion terms decrease by factors $X^{-2/3}/\log X$, but the terms themselves need not, because the leading part of the k -th term vanishes when $3 \mid k$. That sentence and the matching word in Question 4 of Section 24.1 are corrected where they stand, and the proof now derives the leading part and the third and fourth terms; the dated note after the

proof keeps the first wording, gives the counterexample, and records that the remainder in (i), correctly bounded, is numerically one logarithm smaller than its bound. Two optional refinements were adopted: Remark 19.1(b) now gives the shifted grading $\log H(z, u) = b_0 + \sum_{k \geq 1} u^k (b_k + c_{k-1})$, which does reproduce Part II, and Remark 27.3 notes that its first bound is asymptotically sharp. No other statement or number changed; the check found no counterexample to any other statement and no gap in any proof chain.

For Remark 19.1 the check rederived each step and tested it in exact rational series arithmetic: $I''_{m,n} = mn e^z I_{m-1,n-1}$ with $I_{m,n}(0) = I'_{m,n}(0) = 0$ for $1 \leq m, n \leq 4$ through z^{13} ; Elizalde's two recurrences for $k = 1, \dots, 4$; $\sum_k (b_k + c_k) = T$ through z^{15} , with T from the exact recurrence (80); and the shifted grading against $L_n(u)$ of (39) through z^{15} , where the unshifted grading fails already at z^2 . Remark 20.1 was checked by rederivation: the compactness of K_1 , the contour and chord length, the modulus bound with its constant $M(\rho + \varepsilon_1)/(\pi r)$, the expansion of r^{-n} and the ratio $O(e^{-c_1 n})$ to the main term. For Proposition 23.2, with mpmath at 60 digits, the exact x_K found by root-finding, $K = 1, 3$ and $X = 10^3, \dots, 10^{10}$, the error of the truncation (i) divided by $X^{-1}(\log X)^{-3}$ stays in $[0.44, 0.86]$; the error of the displayed formula divided by $X^{-2/3}/\log X$ stays in $[-2.34, -2.11]$ for $K = 3$ and $[-1.33, -1.11]$ for $K = 1$; and the $k = 1, 2$ Lagrange terms, computed by nested numerical differentiation, equal the printed groups to 10^{-30} . The intake comparison in the write note after that proposition was replicated ($\lceil x_3 \rceil = n$ for $n = 100, \dots, 600$, ratio in $[-2.353, -2.333]$; the note's "difference" is the displayed formula minus x_3); the third and fourth terms were evaluated for $X = 10^6, \dots, 10^{20}$, with the values given in the dated note after the proof. For Remark 27.3 both identities for L hold to 10^{-25} on a logarithmic grid $10^{-6} \leq x \leq 316$; the ratio of B to $2e^{-x}(1+q)^{-3}$ is below 1 everywhere, one minus it being 0.950 at $x = 10^{-6}$, 0.568 at $x = 1$, 3.3×10^{-8} at $x = 20$ and 4.0×10^{-25} at $x = 60$; and $\int_0^\infty B = 0.144293678991 \dots$

For Part IV, a word-level comparison of its text (write notes and the [write] remark removed, the label prefix stripped) with the delivered manuscript showed only the declared editorial changes: the abstract printed as a quotation, the added section labels, the reference to Report 87 printed as [13], and $\mathcal{C}_k(r)$ in all three places. The counts recomputed from (80)–(81) equal the 601 terms of `data/06-diagonal-counts_600.json`. Evaluating $A(z)$ from the global branch $F(e^{z/2})$ of (98), independently of the shipped script (which uses the local formula), the check confirmed the bank formula (103) for $k \in \{-1, 0, 1, 2, 3\}$ and $x \in \{0.3, 2, 7\}$, the local branch (100) for $k \in \{-3, 0, 1, 2\}$, the period $4\pi i$ (and that $2\pi i$ is not a period), and the decomposition (107) at $r = 0.5$: the relative error of $\sum_{|k| \leq 6} H_k(n)$ is 3.5×10^{-12} , 3.5×10^{-26} and 1.2×10^{-35} at $n = 6, 15$ and 30 (at $n = 6$ it is 5.8×10^{-17} with $|k| \leq 30$), within the rectangle bound (111) wherever that bound lies above the working precision; $H_{-1} = \overline{H_1}$, H_0 is real, and $H_0/b_{30} - 1 = -7.5175154 \times 10^{-22} = -2 \operatorname{Re} H_1/b_{30}$, as the write note after Corollary 28.2 states. The printed constants $|\zeta_1|$, $|\zeta_2|$, κ_1 , C_1 , $c_{1,1}$, κ_2 , C_2 , $c_{1,2}$, $c_{2,1}$, $\rho/|\zeta_1|$ and $c - \operatorname{Re} \kappa_1$ agree to every printed digit, and $g_k \pi / \sqrt{\zeta_k \kappa_k} = 2\kappa_k$ holds on the principal branches for $k = 0, 1, 2$. A SymPy rebuild of $S(h, y)$ from the definitions (120)–(121) through h^6 cancels the negative and zero powers, reproduces (129) and (130), and by Gaussian moments gives $c_{k,1}$ and $c_{k,2}$ exactly as printed; $c_{0,2}$ is Part I's a_2 , and $c_{0,3}$ is Part III's c_3 . Finally, quadrature of the global branch (banks to 30) reproduces all six entries $Q_k(n)$ of the table in Section 32 to the 12 printed digits (differences at most 5×10^{-13}) and the scaled remainders to 4×10^{-9} . A first run with coarse breakpoints for the oscillating bank integrand disagreed at 10^{-10} ; refined breakpoints, at 30, 50 and 60 digits and with both forms of the integrand, give one value, the printed one, so the discrepancy was the quadrature's. As in the source, these are numerical regressions, not interval certificates. This was a careful reading with numerical tests, not a formal verification or an external review.

A Third and fourth correction coefficients

For convenient exact reproduction, define

$$P_3 = 5832c^9(\rho - 1)^3 + c^6(98172\rho^2 - 173016\rho + 28188) \\ + c^3(134190\rho + 1890) + 665.$$

Then

$$a_3 = -\frac{P_3}{279936c^3}. \quad (132)$$

Similarly let

$$P_4 = 524880c^{12}(\rho - 1)^4 \\ + c^9(17379360\rho^3 - 56337120\rho^2 + 43740000\rho - 4782240) \\ + c^6(84480408\rho^2 - 181744560\rho - 3139560) \\ + c^3(15661800\rho - 693000) + 48125.$$

Then

$$a_4 = \frac{P_4}{201553920c^4}. \quad (133)$$

These expressions are the symbolic output of (28)–(30), rather than fitted constants.

[Write note, 5 October 2026, batch 101.] This appendix belongs to Part I. Its a_3 is also the c_3 of Part III, computed independently by `code/06-diagonal-check_saddle.py` (`data/06-diagonal-saddle_checks.json`, written with a for c).

B Dependency and reproducibility details (Report 94)

[Write note, 5 October 2026, batch 101.] This is Report 94’s Appendix A, printed as Appendix B of this report.

The frozen input is `Exact_Vincular_Avoidance_Asymptotics_Source.zip`, delivered as report 87. Its SHA-256 is

```
1e10659694dd56c8e2e8bf97bb443cb40c1c86e694a363e44295892d1f81c842.
```

The long digest is also provided in machine-readable form in the source bundle. The analytic dependency is exactly (92); the earlier dominant expansion is used only for comparison, not as an unproved input to the new sector theorem.

The tested Python versions are SymPy 1.14.0 and mpmath 1.3.0. A standard pdfL^AT_EX distribution with `amsmath`, `amsthm`, `mathtools`, `geometry`, `lmodern`, `microtype`, `booktabs`, `hyperref`, and `xurl` suffices. Run

```
python -m pip install -r requirements.txt
bash replay.sh
```

The replay uses ordinary symbolic and numerical libraries and performs no external network request. Its mathematical acceptance checks remain active under Python’s optimization flag.

[Write note, 5 October 2026, batch 101.] The archive of Report 87 with this digest arrived in 60f54ea06 and is printed as Part III; its machine-readable record is `data/07-sectors-dependency.json`. Report 94’s `requirements.txt` (the two pins above) is byte-identical to the shipped `data/05-refined-requirements.txt` and is not shipped again. The replay above builds the delivered `article.tex`, which is not shipped, so it runs only in a fresh extraction of the delivered archive; the README gives that route and the commands that run the two checks on a copy of the shipped files. The build script’s TeX paths (`/usr/share/texlive/texmf-dist`, `/usr/share/texmf`) are generic Linux locations, used only when present.

References

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- [10] R. Ferraz de Andrade, E. Lundberg and B. Nagle, *Asymptotics of the extremal excedance set statistic*, arXiv:1403.0691v2 (2016), Eq. (3.1). <https://arxiv.org/abs/1403.0691>. (Part III.)
- [11] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009, Chapter VIII. (Part III; listed but not cited by Report 87.)
- [12] V. Reshetnikov, *ProveIt*, commit 1512ef8356d10fef99e7caaa2753620f0c900c83, *Combinatorial Transseries Inverses*. <https://github.com/VladimirReshetnikov/ProveIt>. (Report 87’s pin; listed but not cited by Report 87.)
- [13] *Exact Vincular Avoidance Asymptotics*, delivered research report 87, 1 October 2026. Frozen source archive and checksum identified in Appendix B. Exact EGF, all-orders dominant expansion, and structural corollaries. (Part IV’s entry; the report is printed as Part III here.)